

FANUC Robot series

**R-30iB CONTROLLER
Local Stop Function with STO**

MAINTENANCE MANUAL

MAREBLSTO002131E REV A

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One or more of the following U.S. patents might be related to the FANUC Robotics products described in this manual.

FANUC Robotics America Corporation Patent List

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Application

This manual describes connection, maintenance and ordering method of the Local Stop function with STO for R-30iB.

It is possible to apply to R-30iB controller of Basic controller, NRTL controller and CE controller with A-cabinet and B-cabinet.

This manual is applicable for both standalone controller and multi arm controller.

Reference

Refer to "Maintenance manual for R-30iB controller(B-83195EN)".

Refer to "R-30iB CONTROLLER AUXILIARY AXIS ORDER MANUAL for B-cabinet (A-95024E)".

Refer to "R-30iB CONTROLLER AUXILIARY AXIS ORDER MANUAL for A-cabinet (A-95023E)".

Refer to "R-30iB Multi arm system Order Manual(A-95025E)".

Refer to "R-30iB CONTROLLER Optional Function OPERATOR'S MANUAL(B-83284EN-2),
AUXILIARY AXIS SERVO OFF (LOCAL STOP) FUNCTION".

Refer to "R-30iB CONTROLLER Dual Check Safety Function OPERATOR'S MANUAL (B-83184EN),
AUXILIARY AXIS SERVO OFF (LOCAL STOP) FUNCTION"
for the setting of safety I/O signals.

Refer to "α i series SERVO AMPLIFIER for 30i-B Descriptions (B-65282EN)".

Safety

FANUC Robotics is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC Robotics therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC Robotics systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC Robotics therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC Robotics training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC Robotics training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.
- Position all controllers outside the robot work envelope.

- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC Robotics recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
- Using a low motion speed, test run the program continuously for at least one full cycle.
- Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

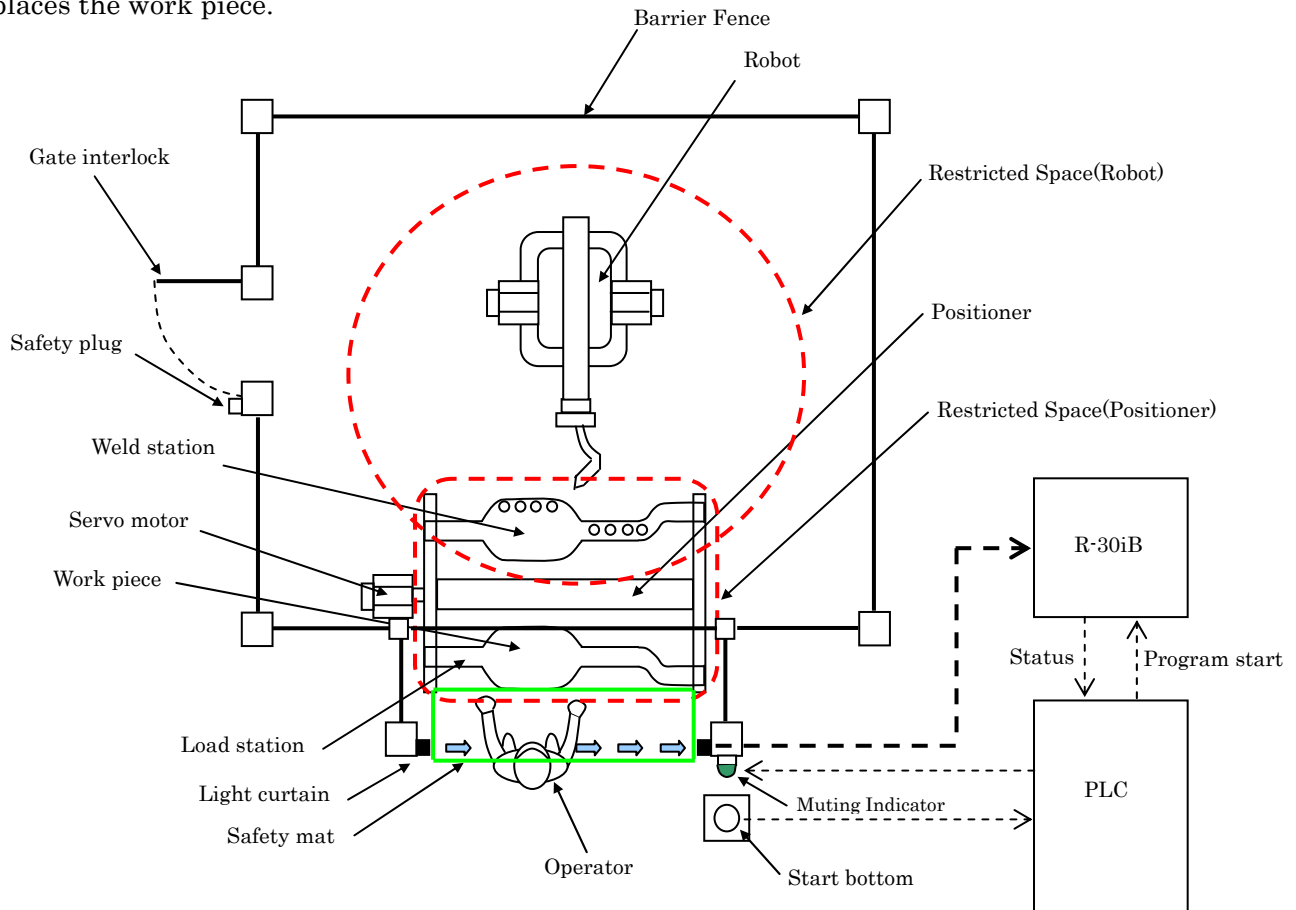
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1 Abstract

1.1 System configurations

The typical work cell with the Local Stop system is shown as follows.
The operator is able to replace the work piece safely while the positioner is stopped in local stop state. And the robot can continue operating(welding, handling) while the operator replaces the work piece.



Feature of local stop function

(1) Safeguarding the operator

When the operator enters in Load station to replace the work piece, the operator's safety is established by disconnecting the power of the positioner with the Local stop function. If the operator enters in the Load station and interrupts the light curtain while energizing the positioner, Emergency Stop will occurs and operator's safety is secured

(2) Improvement of cycle time

When the operator enters in Load station to replace the work piece, it is not necessary to stop the operation of the robot. So the local stop function can improve the cycle time of process.

(3) STO (Safe Torque Off) function

Since Safe Torque Off function based on IEC61800-5-2 is available in the servo amplifier for 30 i-B series CNC, it is possible to cut off the power of the servo amplifier safely. By inputting two interception signals (STO-A, STO-B) into servo amplifier from the outside, PWM signals are cut off in servo amplifier, and the status monitor signal (STO-FB) is outputted. Interception of the power of servo amplifier is safely realizable by monitoring the signal of STO-A, STO-B, and STO-FB.



Warning.

Local stop function apply to the auxiliary axis only(ex. positioner).

Local stop function cannot apply to the robot axis.



Warning.

Safeguarding devices for restricted space

Safeguarding devices(Barrier Fence, Gate Interlock, Safety Plug etc.) are necessary to avoid the operator enters in the restricted space while the robot is operating.



Warning.

Safeguarding devices for load station

Safeguarding devices(light curtain, safety mat etc.) are necessary to generate Emergency Stops when operator enters in the Load station while energizing the positioner.

Safeguarding devices has to cover not only entrance of Load station, but also the whole of Load station by using both light curtain and safety mat as shown in Figure.

See applicable standard for safeguarding requirements.

Confirm the instruction and precautions of safeguarding devices(Light curtain, safety mat etc) before installing the local stop system. And confirm the safety function of safeguarding devices at installation of the local stop system(For example confirm Emergency Stop to occur when the light curtain is interrupted in servo ON).

Terminal of LS brake board for Light curtain is shorted by jumper pin before shipping. So Jumper pin must be removed from terminal and the cable from the safeguarding devices must be connected to the terminal of LS brake board.



Warning.

Safety standard for safeguarding devices

Safeguarding devices	Requirement
Light curtain	Use light curtain conforming to Category 4, PL e (ISO 13849-1), SIL 3 (IEC 61508).
Safety relay	Use safety relay conforming to Category 4, PL e (ISO 13849-1), SIL 3 (IEC 61508).



Warning.

Gravity axis

If the local stop axis has offset loading or is affected by gravity, such that current is required to maintain position, machine lock function(ex. mechanical lock) should be used prior to STO function.



Warning.

Installation of Start Button

In order to prevent the operator from starting the positioner in the load station, the start button that changes from local stop state to servo ON state should be located outside the load station, safety mat and light curtain.



Warning.

Installation of Muting Indicator

The muting indicator which displays the invalidity of the safeguarding devices (light curtain, safety mat etc.) should be located where operator can recognize from outside of the load station in order to prevent the operator from entering to the load station while the positioner is moving.



Warning.

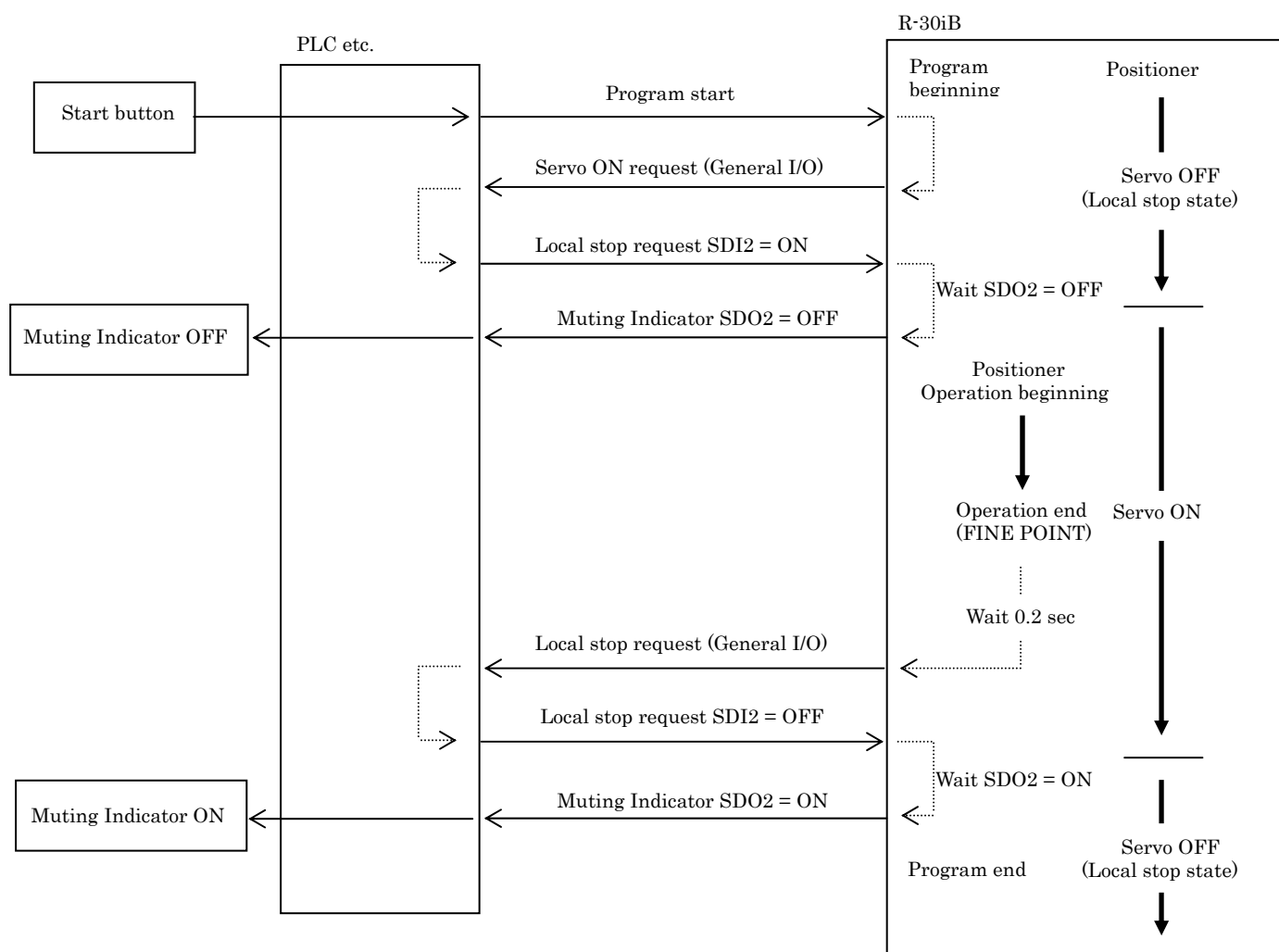
Servo off stop type

The stop type of robot in case that the Light curtain is interrupted in servo ON is defined Controlled stop before shipping. And robot does not stop in T1/T2 mode by default setting.

It is possible to change these setting if necessary. Refer to " R-30iB CONTROLLER Dual Check Safety Function OPERATOR'S MANUAL (B-83184EN), AUXILIARY AXIS SERVO OFF (LOCAL STOP) FUNCTION" .

1.2 Flow chart of local stop system

- (1) The operator enters in the Load station after confirming the Muting indicator is ON, and replaces the work piece. After replacement of work piece, the operator goes out of the Load station, and pushes the start button.
- (2) The positioner program is started by the start button, and the state of positioner is changed from local stop state to servo ON state by local stop request. In the servo ON state the safeguarding (light curtain) are effective. In servo ON state, when the operator enters in the Load station, the light curtain is interrupted and Emergency Stop will occur.
- (3) The operator can recognize the prohibition to enter to Load station according to the Muting indicator. The Muting indicator is controlled by PLC.
- (4) After the positioner program is finished, the state of positioner is changed from servo ON state to local stop state by local stop request. When local stop request is asserted while the positioner is moving, the robot causes the alarm. In the local stop state, the power of positioner is disconnected by local stop function, and the safeguarding (light curtain) is muted. So the operator is able to enter in the Load station to replace the work piece.
- (5) The operator can recognize the permission to enter to Load station according to the Muting indicator.



State of positioner	Robot	Positioner	Safeguarding Device	Enter to Load Station	Muting Indicator
Servo ON	It is possible to operate.	Servo ON	Enable (note 1)	Prohibition	OFF
Servo OFF (Local Stop State)	It is possible to operate.	Servo OFF	Muted (note 2)	Permission	ON

(Note 1) When the light curtain is interrupted in servo ON, the Emergency Stop occurs.

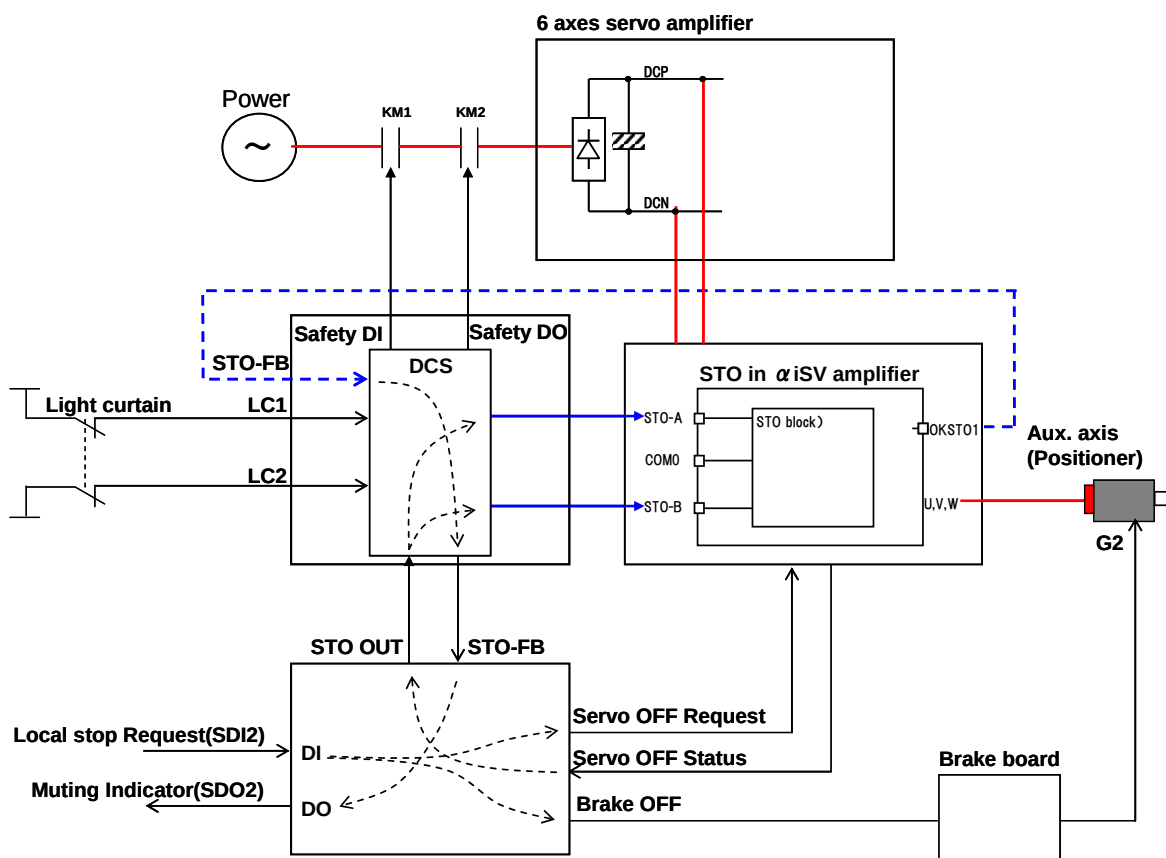
(Note 2) When the light curtain is interrupted in local stop state, Emergency stop does not occur.



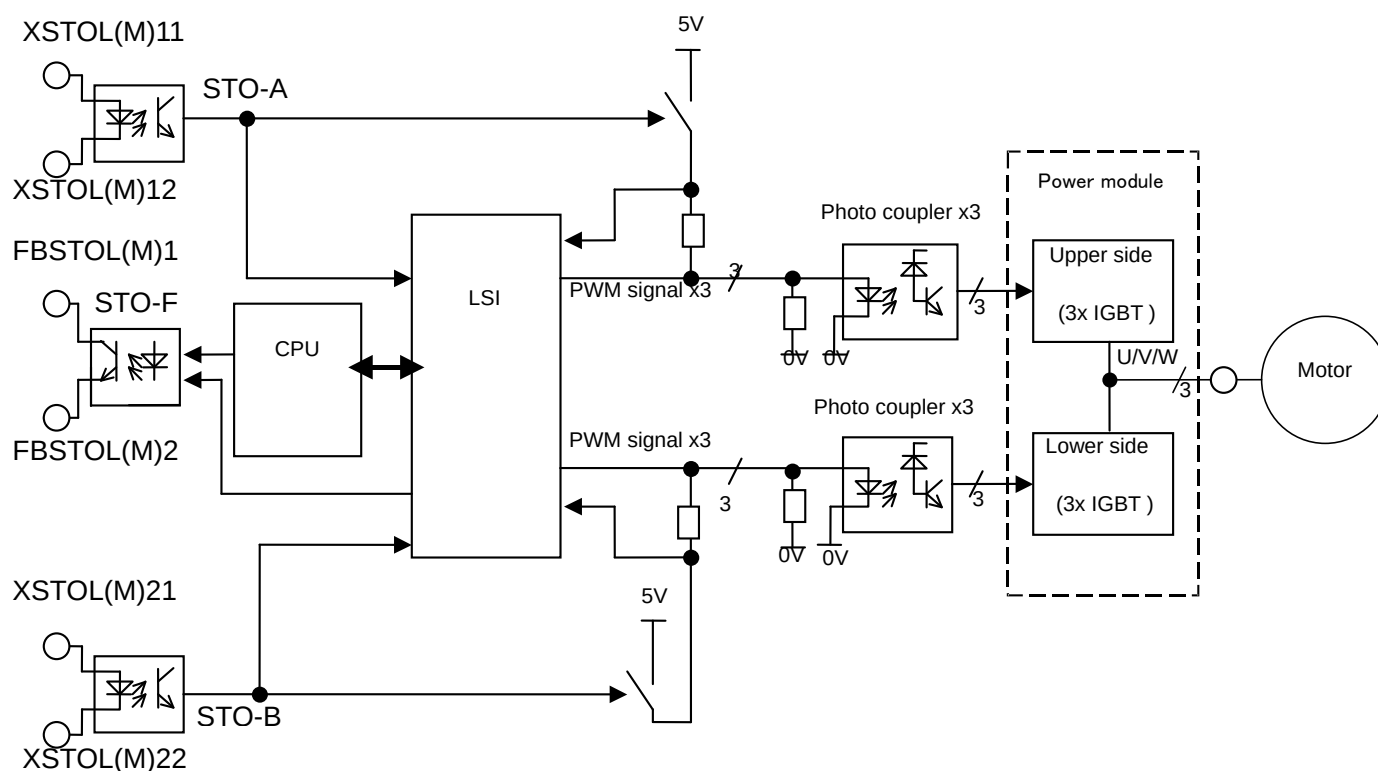
Caution.

The last position before local stop request shall be a “FINE POINT” . It is necessary to keep time more than 0.2 sec before local stop request.

1.3 Block diagram of local stop system with STO



Block diagram of LS with STO



Block diagram of safe torque-off (STO) circuit in α iSV amplifier

1.4 Timing chart of local stop system

The meaning of signals and the timing chart for local stop function are shown as follows.

(1) STO-FB

STO-FB is a feed back signal from servo amplifier.

STO-FB signals of each local stop axes are connected to the Safety DI and DCS software monitors the status of local stop function by this signals.

(2) STO-OUT(STO-A, STO-B)

STO-OUT is a control signal of STO function of servo amplifier. Safe torque off state is valid after turning off the STO-OUT signals.

STO-OUT signals of each local stop axes are connected to the Safety DO. DCS software controls the local stop function by this signals.

(3) Local stop request signal (SDI2)

Local stop request signal(SDI2) is a request signal to transition to local stop state.

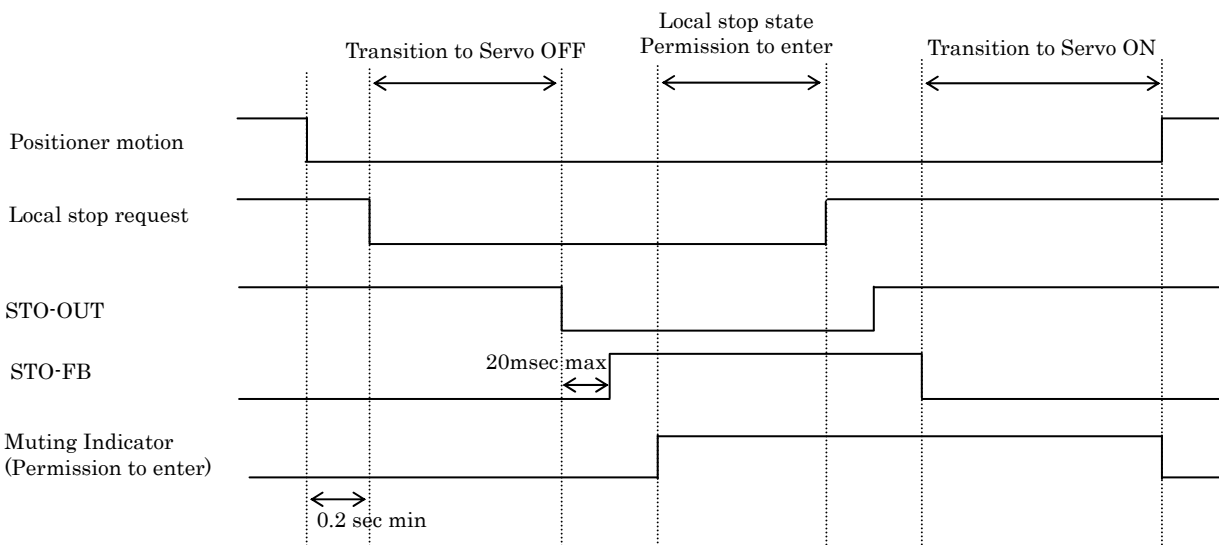
If local stop request signal is asserted when positioner is working, alarm is occurred.

So if the output signals of safeguarding (light curtain, safety mat) are used as the local stop request (SDI2) directly, there is possibility to cause alarm.

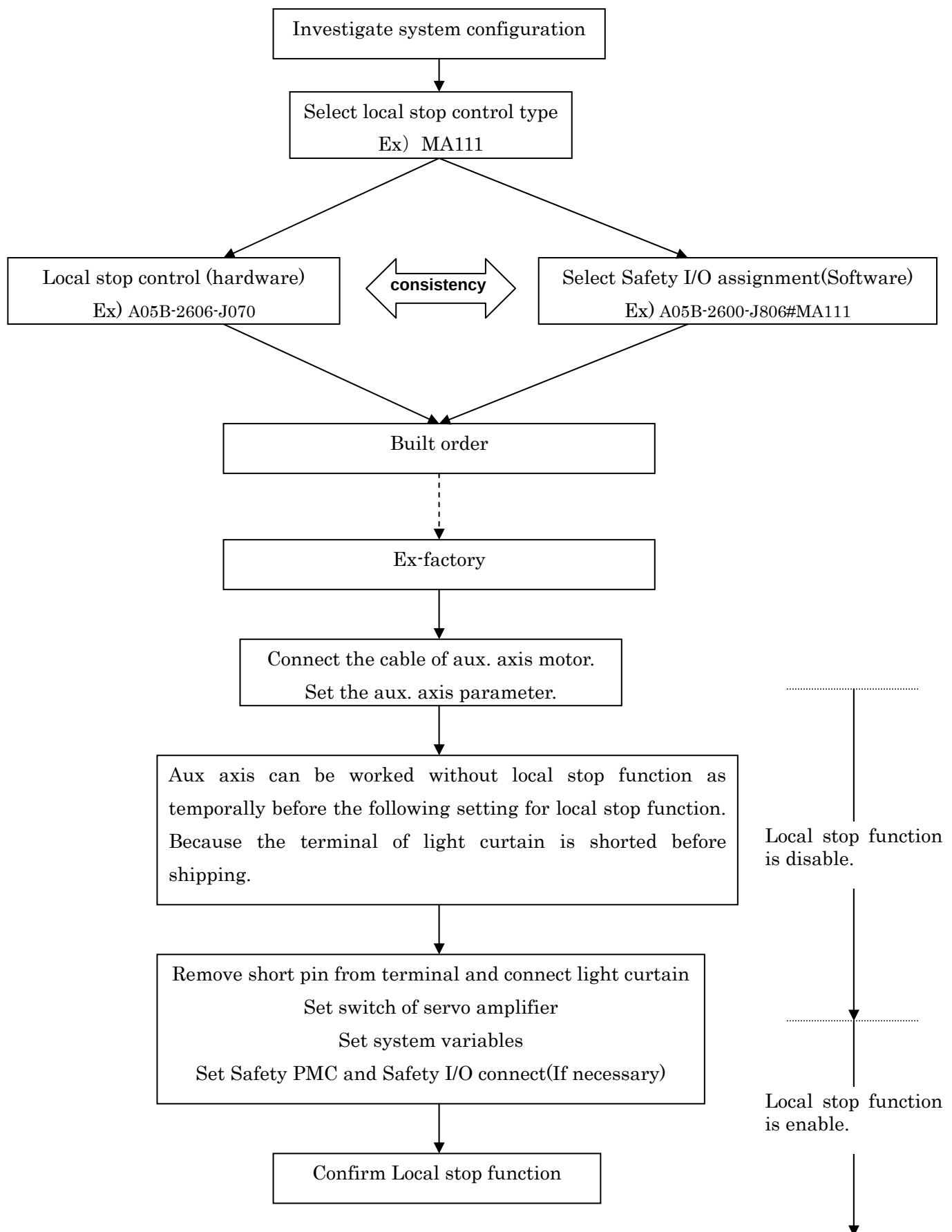
(4) Muting indicator (SDO2)

Muting indicator is a signal to display the status of local stop function.

If operator enters in the Load station when muting indicator is off, it causes the Emergency stop.



1.5 Flow of creating ordering list and system construction



2 Application

2.1 Applicable motors

The applicable motors of local stop function are shown as following table.

Motor	Motor specification	Remarks
αiF 4/4000	A06B-0223-B605	With DC90V brake Taper shaft/
	A06B-0223-B705	With DC90V brake straight shaft/
	A06B-0223-B805	With DC90V brake straight shaft/with key
αiF 8/3000	A06B-0227-B605	With DC90V brake Taper shaft/
	A06B-0227-B705	With DC90V brake straight shaft/
	A06B-0227-B805	With DC90V brake straight shaft/with key
αiF 12/3000	A06B-0243-B605	With DC90V brake Taper shaft/
	A06B-0243-B705	With DC90V brake straight shaft/
	A06B-0243-B805	With DC90V brake straight shaft/with key
αiF 22/3000	A06B-0247-B605	With DC90V brake Taper shaft/
	A06B-0247-B705	With DC90V brake straight shaft/
	A06B-0247-B805	With DC90V brake straight shaft/with key
αiF 40/3000	A06B-0257-B605	With DC90V brake Taper shaft/
	A06B-0257-B705	With DC90V brake straight shaft/
	A06B-0257-B805	With DC90V brake straight shaft/with key
αiS 8/4000	A06B-0235-B605	With DC90V brake Taper shaft/
	A06B-0235-B705	With DC90V brake straight shaft/
	A06B-0235-B805	With DC90V brake straight shaft/with key
αiS 12/4000	A06B-0238-B605	With DC90V brake Taper shaft/
	A06B-0238-B705	With DC90V brake straight shaft/
	A06B-0238-B805	With DC90V brake straight shaft/with key
αiS 22/4000	A06B-0265-B605	With DC90V brake Taper shaft/
	A06B-0265-B705	With DC90V brake straight shaft/
	A06B-0265-B805	With DC90V brake straight shaft/with key
αiS 30/4000	A06B-0268-B605	With DC90V brake Taper shaft/
	A06B-0268-B705	With DC90V brake straight shaft/
	A06B-0268-B805	With DC90V brake straight shaft/with key
αiS 40/4000	A06B-0272-B605	With DC90V brake Taper shaft/
	A06B-0272-B705	With DC90V brake straight shaft/
	A06B-0272-B805	With DC90V brake straight shaft/with key

Motor	Motor specification	Remarks
βiS 8/3000	A06B-0075-B604	With DC90V brake Taper shaft/
	A06B-0075-B704	With DC90V brake straight shaft/
	A06B-0075-B804	With DC90V brake straight shaft/with key
βiS 12/3000	A06B-0078-B604	With DC90V brake Taper shaft/
	A06B-0078-B704	With DC90V brake straight shaft/
	A06B-0078-B804	With DC90V brake straight shaft/with key
βiS 22/2000	A06B-0085-B604	With DC90V brake Taper shaft/
	A06B-0085-B704	With DC90V brake straight shaft/
	A06B-0085-B804	With DC90V brake straight shaft/with key

Note) The following motors can be used in R-30iB controller.

If the following motors will be used in the local stop system, the brake power circuit of local stop function has to be changed. In this case, contact to design section.

αiF1/5000、αiF2/5000、αiS2/5000、αiS4/5000

βiS0.4/5000、βiS0.5/6000、βiS1/6000、βiS2/4000、βiS4/4000

2.2 Applicable servo amplifier

The servo amplifier corresponding to Safe Torque Off (STO) function is shown in following Table.

Edition of servo amplifier that can be used Safe Torque Off (STO) function

Specification	The number of edition
A06B-6240-H101~H106	After edition "B"
A06B-6240-H109	After edition "A"
A06B-6240-H201~H211	After edition "B"
A06B-6240-H301~H308 (*)	After edition "B"
A06B-6290-H101~H105	After edition "B"
A06B-6290-H106~H110	After edition "A"
A06B-6290-H201~H209	After edition "B"
A06B-6290-H301~H305 (*)	After edition "B"



(*) Safe Torque Off (STO) function is invalid for N axis of 3-axis amplifier.

2.3 Aux. axis cabinet

Aux. axis cabinet is out of scope for this manual.

So if it is necessary to use local stop function with aux. axis cabinet, contact to design section.

2.4 Safety I/O board

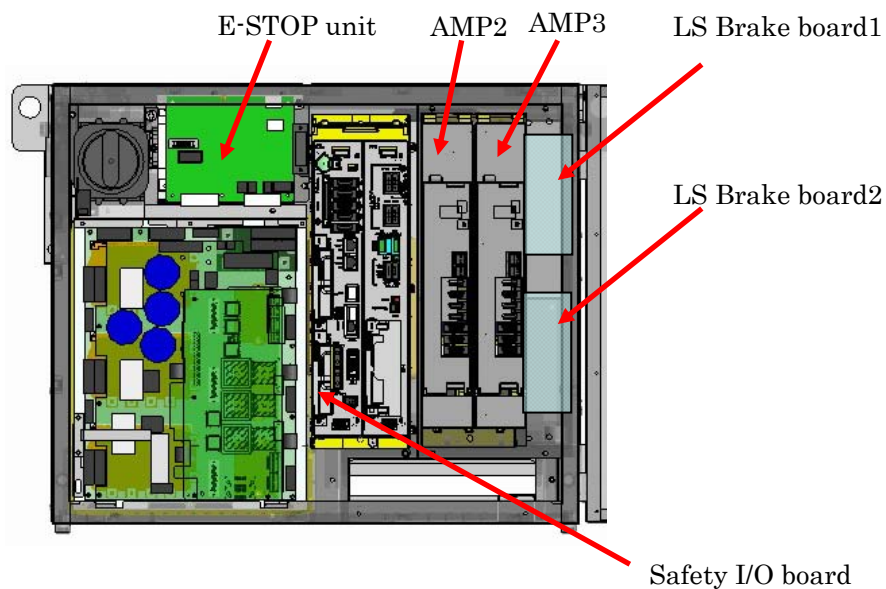
If (2) or more Safety I/O boards are required in R-30iB controller, 1st safety I/O board should be used for local stop function.

3 Mounting and connection

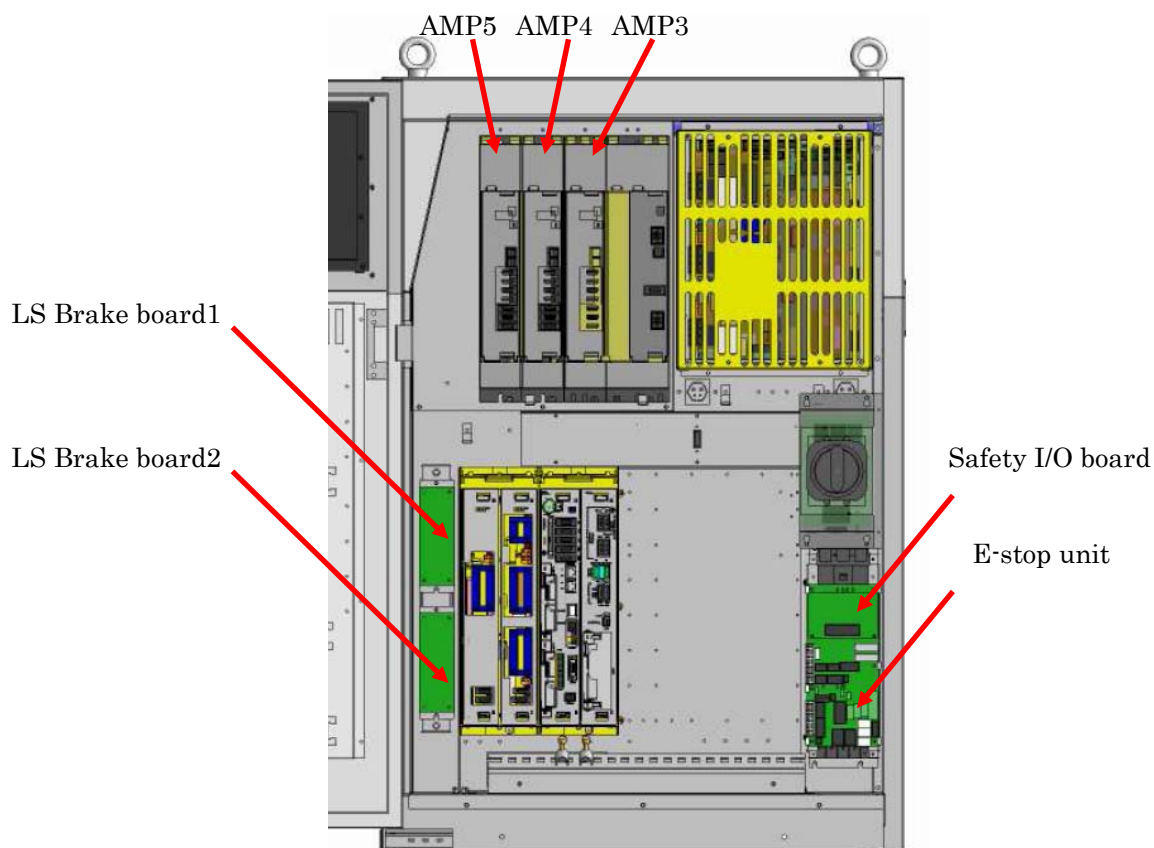
3.1 Mounting diagram

The internal layout of local stop function with STO is shown as follows.

(1) In case of A-cabinet



(2) In case of B-cabinet(with Power regeneration unit)

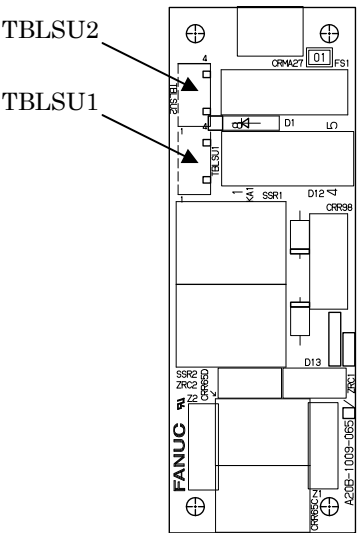


3.2 Connection of Light curtain

LS brake board includes the (2) terminals(TBLSU1,TBLSU2) to connect the cable of light curtain. These terminals are shorted by Jumper pin before shipping. So jumper pin must be removed and the connection of light curtain must be connected to the terminals.

Jumper pin must be removed from “Not Used” terminals(TBLSU1,TBLSU2) such as no connection to the light curtain.

The connection of light curtain signal



LS Brake board

TBLSU1

1	24E
2	LC11
3	0V
4	LC21

Light curtain 1

TBLSU2

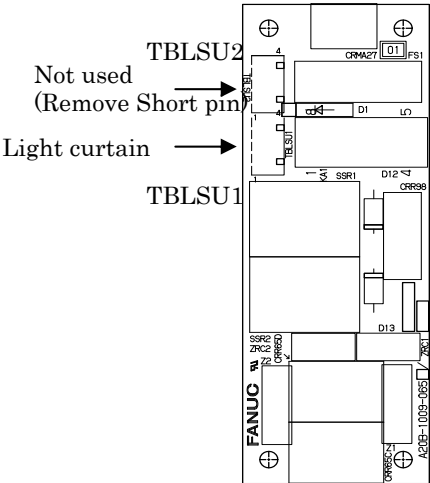
1	24E
2	LC11
3	0V
4	LC21

Light curtain 2

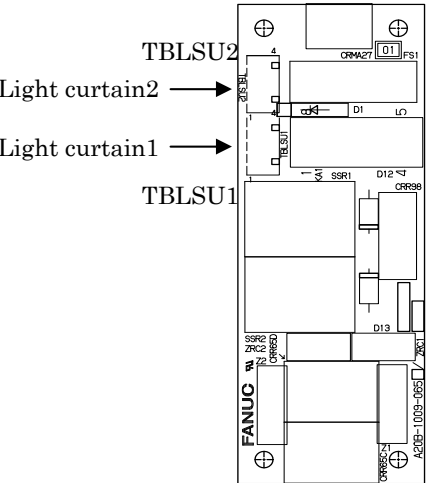
Conductor diameter: 1.25mm² or less
Strip length: 8mm-9mm

Example of Light curtain connection

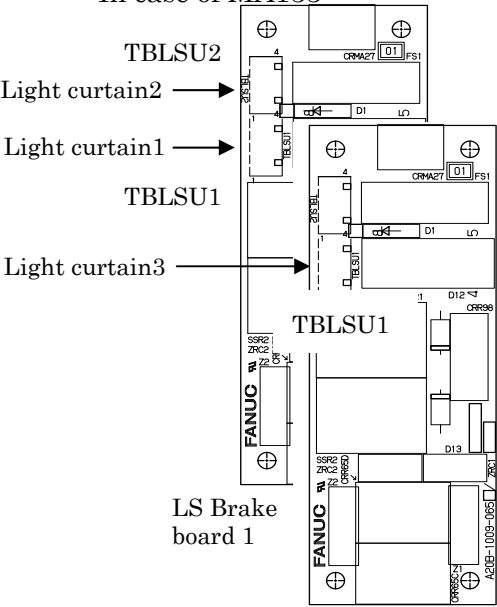
In case of MA111



In case of MA122



In case of MA133



Note) The SFDI (Safety DI) for LC11 and LC21 is reverse polarity.

If the output of light curtain(Polarity, Electrical Level) does not meet the requirement of SFDI, use the safety relay to convert the polarity of light curtain signal.

Connection of light curtain for each configuration

Local stop control type	Master Brake board 1 TBLSU1	Master Brake board 1 TBLSU2	Master Brake board 2 TBLSU1	Master Brake board 2 TBLSU2	Slave Brake board 1 TBLSU1	Slave Brake board 1 TBLSU2
MA111	Positioner1	Not used				
MA111M	Positioner1	Not used				
MA211	Positioner1	Not used				
MA122	Positioner1	Positioner2				
MA222	Positioner1	Positioner2				
MA131	Interchange	Not used	Muting SW1	Muting SW2		
MA131M	Interchange	Not used	Muting SW1	Muting SW2		
MA133	Positioner1	Not used	Positioner2	Positioner3		
MA133M	Positioner1	Not used	Positioner2	Positioner3		
SA122					Positioner1	Positioner2
DA131					Muting SW1	Muting SW2
DA131M					Muting SW1	Muting SW2
DA133					Positioner2	Positioner3
DA133M					Positioner2	Positioner3
MB111	Positioner1	Not used				
MB111M	Positioner1	Not used				
MB211	Positioner1	Not used				
MB122	Positioner1	Positioner2				
MB222	Positioner1	Positioner2				
MB131	Interchange	Not used				
MB131M	Interchange	Not used				
MB133	Positioner1	Not used				
MB133M	Positioner1	Not used				
MB233	Positioner1	Not used				
MB231	Interchange	Not used				
MB231M	Interchange	Not used				

3.3 Setup of the switch of servo amplifier

In order to use Safe Torque Off (STO) function, it is necessary as OFF to set the switch SW1 at the front of servo amplifier. (A default is ON)

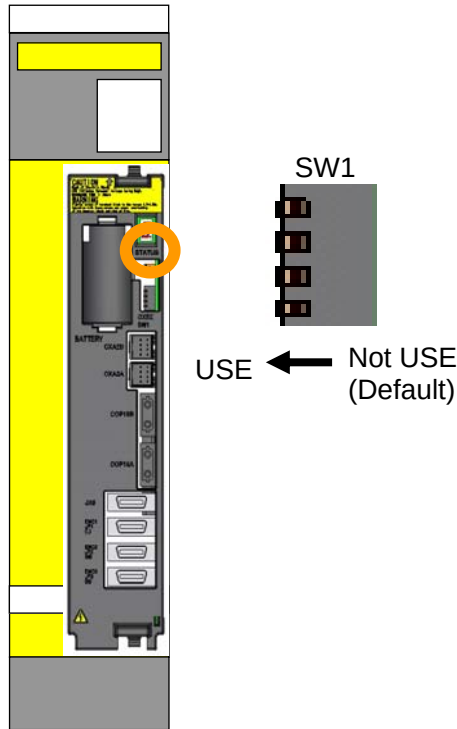
- When you use Safe Torque Off (STO) function in L axis, please turn OFF SW 1-1 and SW 1-2.
- When you use Safe Torque Off (STO) function in M axis, please turn OFF SW 1-3 and SW 1-4.
- Safe Torque Off (STO) function cannot be used with N axis of 3 axes amplifier.

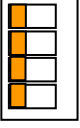
System variable (\$LS_IOPORT[x].\$MO_GRP_NUM) for Local stop function should be changed according to switch setting of servo amplifier. If system variable are not matched switch setting, “SRVO-450 Drvoff circuit alarm” will be occurred.

Setup of system variable

System variable	Value	Status
\$LS_IOPORT[x].\$MO_GRP_NUM	0	Not use(Default)
\$LS_IOPORT[x].\$MO_GRP_NUM	Motion Group number	Use

Setup of switch setting



	L axis 1 st axis	M axis 2 nd axis	N axis 3 rd axis	SW1 setting
1 axis amp				
2 axis amp	Not Use	Not Use	N/A	
3 axis amp				
1 axis amp				
2 axis amp	Use	Not Use	N/A	
3 axis amp				
1 axis amp	Not Use	Use	N/A	
2 axis amp				
3 axis amp	Use	Use	N/A	

4 System configuration and setting

4.1 System configurations

The configurations of the local stop system that can be applied are shown as follows.

In case of A-cabinet

Application	Local stop control type	Axis	Group	Light curtain	Cabinet	Standalone/ Dual arm
1 station(1 axis)	MA111 MA111M	1	1	1	A	Standalone
1 station(2 axes)	MA211	2	1	1	A	Standalone
2 stations(1 axis)	MA122	1	2	2	A	Standalone
2 stations(1 axis) with Rail	MA122	1	2	2	A	Standalone
2 stations(1 axis) with Servo Torch	MA122	1	2	2	A	Standalone
Ferries wheel(1 axis)	MA131 MA131M	1	3	1	A	Standalone
3 stations(1 axis)	MA133 MA133M	1	3	3	A	Standalone
2 stations(2 axes)	MA222	2	2	2	A	Standalone
2 stations(2 axes) with Rail	MA222	2	2	2	A	Standalone
Dual arm + 2 stations(1 axis)	SA122	1	2	2	A	Dual arm
Dual arm + 2 stations(1 axis) + 2 Servo Torch(or 2 Rail)	SA122	1	2	2	A	Dual arm
Dual arm + Ferries wheel(1 axis)	DA131 (MA111+SA122) DA131M (MA111M+SA122)	1	3	1	A	Dual arm
Dual arm + 3 stations(1 axis)	DA133 (MA111+SA122) DA133M (MA111M+SA122)	1	3	3	A	Dual arm

In case of B-cabinet

Application	Local stop control type	Axis	Group	Light curtain	Cabinet	Standalone/ Dual arm
1 station(1 axis)	MB111 MB111M	1	1	1	B	Standalone
1 station(2 axes)	MB211	2	1	1	B	Standalone
2 stations(1 axis)	MB122	1	2	2	B	Standalone
2 stations(1 axis) with Rail	MB122	1	2	2	B	Standalone
2 stations(1 axis) with Servo Torch	MB122	1	2	2	B	Standalone
Ferries wheel(1 axis)	MB131 MB131M	1	3	1	B	Standalone
3 stations(1 axis)	MB133 MB133M	1	3	3	B	Standalone
2 stations(2 axes)	MB222	2	2	2	B	Standalone
2 stations(2 axes) with Rail	MB222	2	2	2	B	Standalone
3 stations(2 axes)	MB233	2	3	3	B	Standalone
Ferries wheel(2 axes)	MB231 MB231M	2	3	1	B	Standalone
Dual arm + 2 stations(1 axis)	MB122	1	2	2	B	Standalone
Dual arm + 2 stations(1 axis)+ 2 Servo Torch (or 2 Rail)	MB122	1	2	2	B	Standalone
Dual arm + Ferries wheel(1 axis)	MB131	1	3	1	B	Standalone
Dual arm + 3 stations(1 axis)	MB133	1	3	3	B	Standalone

4.2 Local stop Control type

The following table is local stop control type.

Local stop control type

Local stop control type	Master /Slave	Cabinet	Axis	Group	Total axis	Light Curtain	Brake Board	Safety I/O Board	AMP number	Brake Number
MA111	Master	A cab	1	1	1	1	1	-	AMP2(L) or AMP3(L)	2
MA111M	Master	A cab	1	1	1	1	1	-	AMP2(M) or AMP3(M)	2
MA211	Master	A cab	2	1	2	1	1	-	AMP2(L and M) or AMP3(L and M)	2
MA122	Master	A cab	1	2	2	2	1	1	AMP2(L and M) or AMP3(L and M)	2,3
MA222	Master	A cab	2	2	4	2	1	1	AMP2(L and M) and AMP3(L and M)	2,3
MA131	Master	A cab	1	3	3	1	2	1	AMP2(L) and AMP3(L and M)	2,3,4
MA131M	Master	A cab	1	3	3	1	2	1	AMP2(M) and AMP3(L and M)	2,3,4
MA133	Master	A cab	1	3	3	3	2	1	AMP2(L) and AMP3(L and M)	2,3,4
MA133M	Master	A cab	1	3	3	3	2	1	AMP2(M) and AMP3(L and M)	2,3,4
SA122	Slave	A cab	1	2	2	2	1	-	AMP2(L and M) or AMP3(L and M)	6,7
MB111	Master	B cab	1	1	1	1	1	1	AMPn(L)	2
MB111M	Master	B cab	1	1	1	1	1	1	AMPn(M)	2
MB211	Master	B cab	2	1	2	1	1	1	AMPn(L or M)	2
MB122	Master	B cab	1	2	2	2	1	1	AMPn(L and M)	2,3
MB222	Master	B cab	2	2	4	2	1	1	AMPn(L and M) and AMPn+1(L and M)	2,3
MB131	Master	B cab	1	3	3	1	2	1	AMPn(L) and AMPn+1(L and M)	2,3,4
MB131M	Master	B cab	1	3	3	1	2	1	AMPn(M) and AMPn+1(L and M)	2,3,4
MB133	Master	B cab	1	3	3	3	2	1	AMPn(L) and AMPn+1(L and M)	2,3,4
MB133M	Master	B cab	1	3	3	3	2	1	AMPn(M) and AMPn+1(L and M)	2,3,4
MB233	Master	B cab	2	3	6	3	2	1	AMPn(L and M) and AMPn+1(L and M) and AMPn+2(L and M)	2,3,4
MB231	Master	B cab	2	3	5	1	2	1	AMPn(L) and AMPn+1(L and M) and AMPn+2(L and M)	2,3,4
MB231M	Master	B cab	2	3	5	1	2	1	AMPn(M) and AMPn+1(L and M) and AMPn+2(L and M)	2,3,4

4.3 I/O Setting of each local stop control type.

The assignment of Safety I/O signals to control local stop function is shown as follows.

Safety I/O setting table for local stop function with STO

TYPE	Soft DI	Hard DI	FENCE/FB	Soft DO	Hard DO	AMP	STO-OUT	Axis	Brake
MA111 MA111M	SPI1+	SFDI11	LC11	SPO1+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
						AMPn(M)	STO-A2		
	SPI1-	SFDI21	LC21	SPO1-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
	SPI2+	SFDI12	STO-FB1	-	-	-	-	-	-
	SPI2-	SFDI22	STO-FB2	-	-	-	-	-	-
MA211	SPI1+	SFDI11	LC11	SPO1+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
						AMPn(M)	STO-A2		
	SPI1-	SFDI21	LC21	SPO1-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
	SPI2+	SFDI12	STO-FB1	-	-	-	-	-	-
	SPI2-	SFDI22	STO-FB2	-	-	-	-	-	-
MA122	SPI9+	SFDI11	LC11	SPO9+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
	SPI9-	SFDI21	LC21	SPO9-	SFDO21		STO-B1		
	SPI10+	SFDI12	LC12	SPO10+	SFDO12	AMPn(M)	STO-A2	Positioner2	3
	SPI10-	SFDI22	LC22	SPO10-	SFDO22		STO-B2		
	SPI11+	SFDI13	STO-FB1	-	-	-	-	-	-
	SPI11-	SFDI23	STO-FB2	-	-	-	-	-	-
MA222	SPI9+	SFDI11	LC11	SPO9+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
						AMPn(M)	STO-A2		
	SPI9-	SFDI21	LC21	SPO9-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
	SPI10+	SFDI12	LC12	SPO10+	SFDO12	AMPn+1(L)	STO-A1	Positioner2	3
						AMPn+1(M)	STO-A2		
	SPI10-	SFDI22	LC22	SPO10-	SFDO22	AMPn+1(L)	STO-B1		
						AMPn+1(M)	STO-B2		
	SPI11+	SFDI13	STO-FB1	-	-	-	-	-	-
	SPI11-	SFDI23	STO-FB2	-	-	-	-	-	-
	SPI12+	SFDI14	STO-FB3	-	-	-	-	-	-
	SPI12-	SFDI24	STO-FB4	-	-	-	-	-	-
MA131 MA131M	SPI9+	SFDI11	LC11	SPO9+	SFDO11	AMPn(L)	STO-A1	Interchange	2
						AMPn(M)	STO-A2		
	SPI9-	SFDI21	LC21	SPO9-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
	SPI10+	SFDI12	Not Use	-	-	-	-		-
	SPI10-	SFDI22	Not Use	-	-	-	-		-
	SPI11+	SFDI13	MS11	SPO11+	SFDO13	AMPn+1(L)	STO-A1	Positioner1	3
	SPI11-	SFDI23	MS21	SPO11-	SFDO23		STO-B1		
	SPI12+	SFDI14	MS12	SPO12+	SFDO14	AMPn+1(M)	STO-A2	Positioner2	4
	SPI12-	SFDI24	MS22	SPO12-	SFDO24		STO-B2		
	SPI13+	SFDI15	STO-FB1	-	-	-	-		-
	SPI13-	SFDI25	STO-FB2	-	-	-	-		-
	SPI14+	SFDI16	STO-FB3	-	-	-	-		-
	SPI14-	SFDI26	STO-FB4	-	-	-	-		-

TYPE	Soft DI	Hard DI	FENCE/FB	Soft DO	Hard DO	AMP	STO-OUT	Axis	Brake
MA133 MA133M	SPI9+	SFDI11	LC11	SPO9+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
						AMPn(M)	STO-A2		
	SPI9-	SFDI21	LC21	SPO9-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
	SPI10+	SFDI12	Not Use	-	-	-	-	-	-
	SPI10-	SFDI22	Not Use	-	-	-	-	-	-
	SPI11+	SFDI13	LC12	SPO11+	SFDO13	AMPn+1(L)	STO-A1	Positioner1	3
	SPI11-	SFDI23	LC22	SPO11-	SFDO23		STO-B1		
	SPI12+	SFDI14	LC13	SPO12+	SFDO14	AMPn+1(M)	STO-A2	Positioner2	4
	SPI12-	SFDI24	LC23	SPO12-	SFDO24		STO-B2		
	SPI13+	SFDI15	STO-FB1	-	-	-	-	-	-
	SPI13-	SFDI25	STO-FB2	-	-	-	-	-	-
SA122	SPI14+	SFDI16	STO-FB3	-	-	-	-	-	-
	SPI14-	SFDI26	STO-FB4	-	-	-	-	-	-
	SPI49+	SFDI11	LC11	SPO49+	SFDO11	AMPn(L)	STO-A1	Positioner1	6
	SPI49-	SFDI21	LC21	SPO49-	SFDO21		STO-B1		
	SPI50+	SFDI12	LC12	SPO50+	SFDO12	AMPn(M)	STO-A2	Positioner2	7
	SPI50-	SFDI22	LC22	SPO50-	SFDO22		STO-B2		
	SPI58+	FENCE1	STO-FB1	-	-	-	-	-	-
	SPI58-	FENCE2	STO-FB2	-	-	-	-	-	-

TYPE	Soft DI	Hard DI	FENCE/FB	Soft DO	Hard DO	AMP	STO-OUT	Axis	Brake
MB111 MB111M	SPI1+	SFDI11	LC11	SPO1+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
						AMPn(M)	STO-A2		
	SPI1-	SFDI21	LC21	SPO1-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
MB211	SPI2+	SFDI12	STO-FB1	-	-	-	-	-	-
						-	-	-	-
	SPI2-	SFDI22	STO-FB2	-	-	-	-	-	-
						-	-	-	-
MB122	SPI1+	SFDI11	LC11	SPO1+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
	SPI1-	SFDI21	LC21	SPO1-	SFDO21		STO-B1		
	SPI2+	SFDI12	LC12	SPO2+	SFDO12	AMPn(M)	STO-A2	Positioner2	3
	SPI2-	SFDI22	LC22	SPO2-	SFDO22		STO-B2		
	SPI3+	SFDI13	STO-FB1	-	-	-	-	-	-
	SPI3-	SFDI23	STO-FB2	-	-	-	-	-	-
MB222	SPI1+	SFDI11	LC11	SPO1+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
						AMPn(M)	STO-A2		
	SPI1-	SFDI21	LC21	SPO1-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
	SPI2+	SFDI12	LC12	SPO2+	SFDO12	AMPn+1(L)	STO-A1	Positioner2	3
						AMPn+1(M)	STO-A2		
	SPI2-	SFDI22	LC22	SPO2-	SFDO22	AMPn+1(L)	STO-B1		
						AMPn+1(M)	STO-B2		
	SPI3+	SFDI13	STO-FB1	-	-	-	-	-	-
	SPI3-	SFDI23	STO-FB2	-	-	-	-	-	-
	SPI4+	SFDI14	STO-FB3	-	-	-	-	-	-
	SPI4-	SFDI24	STO-FB4	-	-	-	-	-	-

TYPE	Soft DI	Hard DI	FENCE/FB	Soft DO	Hard DO	AMP	STO-OUT	Axis	Brake
MB131 MB131M	SPI1+	SFDI11	LC11	SPO1+	SFDO11	AMPn(L)	STO-A1	Interchange	2
						AMPn(M)	STO-A2		
	SPI1-	SFDI21	LC21	SPO1-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
	SPI2+	SFDI12	Not Use	-	-	-	-	-	-
	SPI2-	SFDI22	Not Use	-	-	-	-	-	-
	SPI3+	SFDI13	MS11	SPO3+	SFDO13	AMPn+1(L)	STO-A1	Positioner1	3
	SPI3-	SFDI23	MS21	SPO3-	SFDO23		STO-B1		
	SPI4+	SFDI14	MS12	SPO4+	SFDO14	AMPn+1(M)	STO-A2	Positioner2	4
	SPI4-	SFDI24	MS22	SPO4-	SFDO24		STO-B2		
	SPI5+	SFDI15	STO-FB1	-	-	-	-	-	-
	SPI5-	SFDI25	STO-FB2	-	-	-	-	-	-
MB133 MB133M	SPI1+	SFDI11	LC11	SPO1+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
						AMPn(M)	STO-A2		
	SPI1-	SFDI21	LC21	SPO1-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
	SPI2+	SFDI12	Not Use	-	-	-	-	-	-
	SPI2-	SFDI22	Not Use	-	-	-	-	-	-
	SPI3+	SFDI13	MS11	SPO3+	SFDO13	AMPn+1(L)	STO-A1	Positioner2	3
	SPI3-	SFDI23	MS21	SPO3-	SFDO23		STO-B1		
	SPI4+	SFDI14	MS12	SPO4+	SFDO14	AMPn+1(M)	STO-A2	Positioner3	4
	SPI4-	SFDI24	MS22	SPO4-	SFDO24		STO-B2		
	SPI5+	SFDI15	STO-FB1	-	-	-	-	-	-
	SPI5-	SFDI25	STO-FB2	-	-	-	-	-	-
MB233	SPI1+	SFDI11	LC11	SPO1+	SFDO11	AMPn(L)	STO-A1	Positioner1	2
							STO-A2		
	SPI1-	SFDI21	LC21	SPO1-	SFDO21	AMPn(M)	STO-B1		
							STO-B2		
	SPI2+	SFDI12	Not Use	-	-	-	-	-	-
	SPI2-	SFDI22	Not Use	-	-	-	-	-	-
	SPI3+	SFDI13	LC12	SPO3+	SFDO13	AMPn+1(L)	STO-A1	Positioner2	3
	SPI3-	SFDI23	LC22	SPO3-	SFDO23	AMPn+1(M)	STO-A2		
							STO-B1		
	SPI4+	SFDI14	LC13	SPO4+	SFDO14	AMPn+2(L)	STO-B2		
							STO-A1	Positioner3	4
	SPI4-	SFDI24	LC23	SPO4-	SFDO24	AMPn+2(M)	STO-A2		
							STO-B1		
	SPI5+	SFDI15	STO-FB1	-	-	-	STO-B2		
							-	-	-
	SPI5-	SFDI25	STO-FB2	-	-	-	-	-	-
	SPI6+	SFDI16	STO-FB3	-	-	-	-	-	-
	SPI6-	SFDI26	STO-FB4	-	-	-	-	-	-

TYPE	Soft DI	Hard DI	FENCE/FB	Soft DO	Hard DO	AMP	STO-OUT	Axis	Brake
MB231 MB231M	SPI1+	SFDI11	LC11	SPO1+	SFDO11	AMPn(L)	STO-A1	Interchange	2
						AMPn(M)	STO-A2		
	SPI1-	SFDI21	LC21	SPO1-	SFDO21	AMPn(L)	STO-B1		
						AMPn(M)	STO-B2		
	SPI2+	SFDI12	Not Use	-	-	-	-	-	-
	SPI2-	SFDI22	Not Use	-	-	-	-	-	-
	SPI3+	SFDI13	MS11	SPO3+	SFDO13	AMPn+1(L)	STO-A1	Positioner1	3
							STO-A2		
	SPI3-	SFDI23	MS21	SPO3-	SFDO23	AMPn+1(M)	STO-B1		
							STO-B2		
	SPI4+	SFDI14	MS12	SPO4+	SFDO14	AMPn+2(L)	STO-A1	Positioner2	4
							STO-A2		
	SPI4-	SFDI24	MS22	SPO4-	SFDO24	AMPn+2(M)	STO-B1		
							STO-B2		
	SPI5+	SFDI15	STO-FB1	-	-	-	-	-	-
	SPI5-	SFDI25	STO-FB2	-	-	-	-	-	-
	SPI6+	SFDI16	STO-FB3	-	-	-	-	-	-
	SPI6-	SFDI26	STO-FB4	-	-	-	-	-	-

4.4 Setting of Safety I/O

The setting of Safety I/O for local stop function with STO is follows.

The assignment of Safety I/O are set before shipping according to local stop control type.

In case of A-cabinet

DCS					
Local Stop					1/10
Stop Type : Controlled stop					OK
Fence in T1/T2 mode : ENABLE					OK
No.	STO-OUT	STO-FB	Fence	Status	
1	@SPO[1]	SPI[2]-1	@SPI[1]	OK	
2	SPO[0]	SPI[0]-1	SPI[0]	OK	
3	SPO[0]	SPI[0]-1	SPI[0]	OK	
4	SPO[0]	SPI[0]-1	SPI[0]	OK	
5	SPO[0]	SPI[0]-1	SPI[0]	OK	
6	SPO[0]	SPI[0]-1	SPI[0]	OK	
7	SPO[0]	SPI[0]-1	SPI[0]	OK	
8	SPO[0]	SPI[0]-1	SPI[0]	OK	
[TYPE]					UNDO

TYPE	No.	STO-OUT	STO-FB	FENCE	\$LS_IOPORT[n].\$SDO1_P_NUM
MA111	1	SPO[1]	SPI[2]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
MA111M	1	SPO[1]	SPI[2]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
MA211	1	SPO[1]	SPI[2]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[1]	SPI[2]-2	SPI[1]	
MA122	1	SPO[9]	SPI[11]-1	SPI[9]	\$LS_IOPORT[1].\$SDO1_P_NUM = 9
	2	SPO[10]	SPI[11]-2	SPI[10]	\$LS_IOPORT[2].\$SDO1_P_NUM = 10
MA222	1	SPO[9]	SPI[11]-1	SPI[9]	\$LS_IOPORT[1].\$SDO1_P_NUM = 9
	2	SPO[9]	SPI[11]-2	SPI[9]	
	3	SPO[10]	SPI[12]-1	SPI[10]	\$LS_IOPORT[2].\$SDO1_P_NUM = 10
	4	SPO[10]	SPI[12]-2	SPI[10]	
MA131	1	SPO[9]	SPI[13]-1	SPI[9]	\$LS_IOPORT[1].\$SDO1_P_NUM = 9
	2	SPO[11]	SPI[14]-1	SPI[9]	\$LS_IOPORT[2].\$SDO1_P_NUM = 11
	3	SPO[12]	SPI[14]-2	SPI[9]	\$LS_IOPORT[3].\$SDO1_P_NUM = 12
MA131M	1	SPO[9]	SPI[13]-2	SPI[9]	\$LS_IOPORT[1].\$SDO1_P_NUM = 9
	2	SPO[11]	SPI[14]-1	SPI[9]	\$LS_IOPORT[2].\$SDO1_P_NUM = 11
	3	SPO[12]	SPI[14]-2	SPI[9]	\$LS_IOPORT[3].\$SDO1_P_NUM = 12
MA133	1	SPO[9]	SPI[13]-1	SPI[9]	\$LS_IOPORT[1].\$SDO1_P_NUM = 9
	2	SPO[11]	SPI[14]-1	SPI[11]	\$LS_IOPORT[2].\$SDO1_P_NUM = 11
	3	SPO[12]	SPI[14]-2	SPI[12]	\$LS_IOPORT[3].\$SDO1_P_NUM = 12
MA133M	1	SPO[9]	SPI[13]-2	SPI[9]	\$LS_IOPORT[1].\$SDO1_P_NUM = 9
	2	SPO[11]	SPI[14]-1	SPI[11]	\$LS_IOPORT[2].\$SDO1_P_NUM = 11
	3	SPO[12]	SPI[14]-2	SPI[12]	\$LS_IOPORT[3].\$SDO1_P_NUM = 12
SA122	1	SPO[49]	SPI[58]-1	SPI[49]	\$LS_IOPORT[1].\$SDO1_P_NUM = 49
	2	SPO[50]	SPI[58]-2	SPI[50]	\$LS_IOPORT[2].\$SDO1_P_NUM = 50

TYPE	No.	STO-OUT	STO-FB	FENCE	\$LS_IOPORT[n].\$SDO1_P_NUM
DA131	1	SPO[1]	SPI[2]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[49]	SPI[58]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 49
	3	SPO[50]	SPI[58]-2	SPI[1]	\$LS_IOPORT[2].\$SDO1_P_NUM = 50
DA131M	1	SPO[1]	SPI[2]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[49]	SPI[58]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 49
	3	SPO[50]	SPI[58]-2	SPI[1]	\$LS_IOPORT[2].\$SDO1_P_NUM = 50
DA133	1	SPO[1]	SPI[2]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[49]	SPI[58]-1	SPI[49]	\$LS_IOPORT[1].\$SDO1_P_NUM = 49
	3	SPO[50]	SPI[58]-2	SPI[50]	\$LS_IOPORT[2].\$SDO1_P_NUM = 50
DA133M	1	SPO[1]	SPI[2]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[49]	SPI[58]-1	SPI[49]	\$LS_IOPORT[1].\$SDO1_P_NUM = 49
	3	SPO[50]	SPI[58]-2	SPI[50]	\$LS_IOPORT[2].\$SDO1_P_NUM = 50

DCS				
Local Stop				1/10
Stop Type : Controlled stop				OK
Fence in T1/T2 mode : ENABLE				OK
No.	STO-OUT	STO-FB	Fence	Status
1	@SPO[1]	SPI[2]-1	@SPI[1]	OK
2	SPO[0]	SPI[0]-1	SPI[0]	OK
3	SPO[0]	SPI[0]-1	SPI[0]	OK
4	SPO[0]	SPI[0]-1	SPI[0]	OK
5	SPO[0]	SPI[0]-1	SPI[0]	OK
6	SPO[0]	SPI[0]-1	SPI[0]	OK
7	SPO[0]	SPI[0]-1	SPI[0]	OK
8	SPO[0]	SPI[0]-1	SPI[0]	OK
[TYPE]				UNDO

In case of B-cabinet

TYPE	No.	STO-OUT	STO-FB	FENCE	\$LS_IOPORT[n].\$SDO1_P_NUM
MB111	1	SPO[1]	SPI[2]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
MB111M	1	SPO[1]	SPI[2]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
MB211	1	SPO[1]	SPI[2]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[1]	SPI[2]-2	SPI[1]	
MB122	1	SPO[1]	SPI[3]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[2]	SPI[3]-2	SPI[2]	\$LS_IOPORT[2].\$SDO1_P_NUM = 2
MB222	1	SPO[1]	SPI[3]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[1]	SPI[3]-2	SPI[1]	
	3	SPO[2]	SPI[4]-1	SPI[2]	\$LS_IOPORT[2].\$SDO1_P_NUM = 2
	4	SPO[2]	SPI[4]-2	SPI[2]	
MB131	1	SPO[1]	SPI[5]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	SPI[1]	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[4]	SPI[6]-2	SPI[1]	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
MB131M	1	SPO[1]	SPI[5]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	SPI[1]	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[4]	SPI[6]-2	SPI[1]	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
MB133	1	SPO[1]	SPI[5]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	SPI[3]	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[4]	SPI[6]-2	SPI[4]	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
MB133M	1	SPO[1]	SPI[5]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	SPI[3]	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[4]	SPI[6]-2	SPI[4]	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
MB233	1	SPO[1]	SPI[5]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[1]	SPI[5]-2	SPI[1]	
	3	SPO[3]	SPI[6]-1	SPI[3]	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	4	SPO[3]	SPI[6]-2	SPI[3]	
	5	SPO[4]	SPI[7]-1	SPI[4]	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
	6	SPO[4]	SPI[7]-2	SPI[4]	
MB231	1	SPO[1]	SPI[5]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	SPI[1]	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[3]	SPI[6]-2	SPI[1]	
	4	SPO[4]	SPI[7]-1	SPI[1]	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
	5	SPO[4]	SPI[7]-2	SPI[1]	
MB231M	1	SPO[1]	SPI[5]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	SPI[1]	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[3]	SPI[6]-2	SPI[1]	
	4	SPO[4]	SPI[7]-1	SPI[1]	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
	5	SPO[4]	SPI[7]-2	SPI[1]	

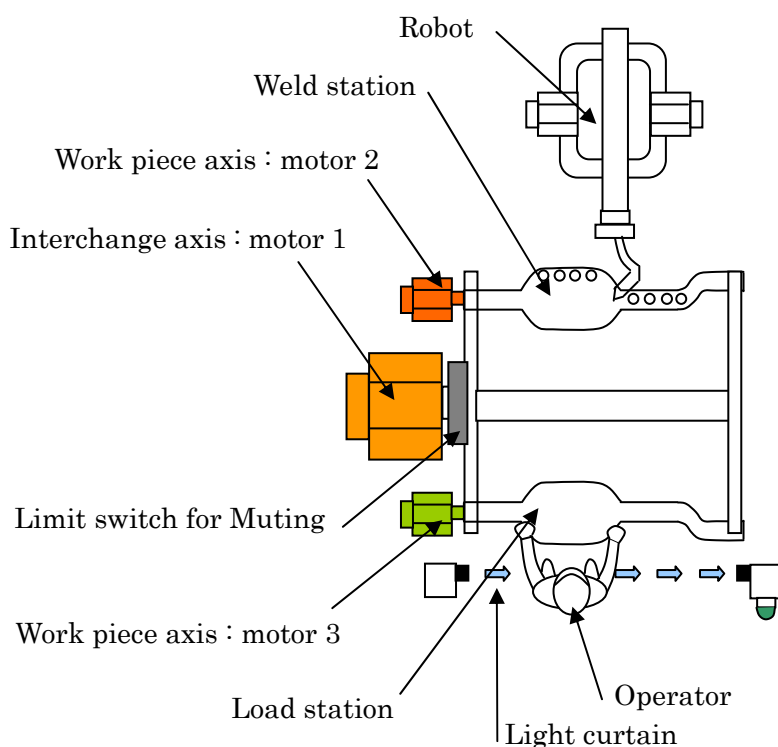
5 Ferris wheel positioner

5.1 System configuration of Ferris wheel positioner

The example of Ferris Wheel positioner and typical Local Stop system with Ferris Wheel positioner(3 axes positioner) are shown as follows. Local stop control types for Ferris wheel system are MA131, MA131M, MB131, MB131M, MB231 and MB231M.

The Ferris Wheel positioner is comprised of two work piece axes(Motor 2, Motor 3) supported by a central interchange axis(Motor 1). One work piece is indexed to a fixed position within the envelope of the robot. The remaining work piece is positioned at the load station. The work piece axes are interchanged between load station and weld station by 180 degree indexing of the interchange axis.

The operator is able to replace the work piece of load station safely while the interchange axis and work piece axis of load station are stopped in local stop state. The rotation of work piece axis at weld station can be coordinated with the robot motion while the operator replaces the work piece of load station side.



The example of Ferris Wheel positioner

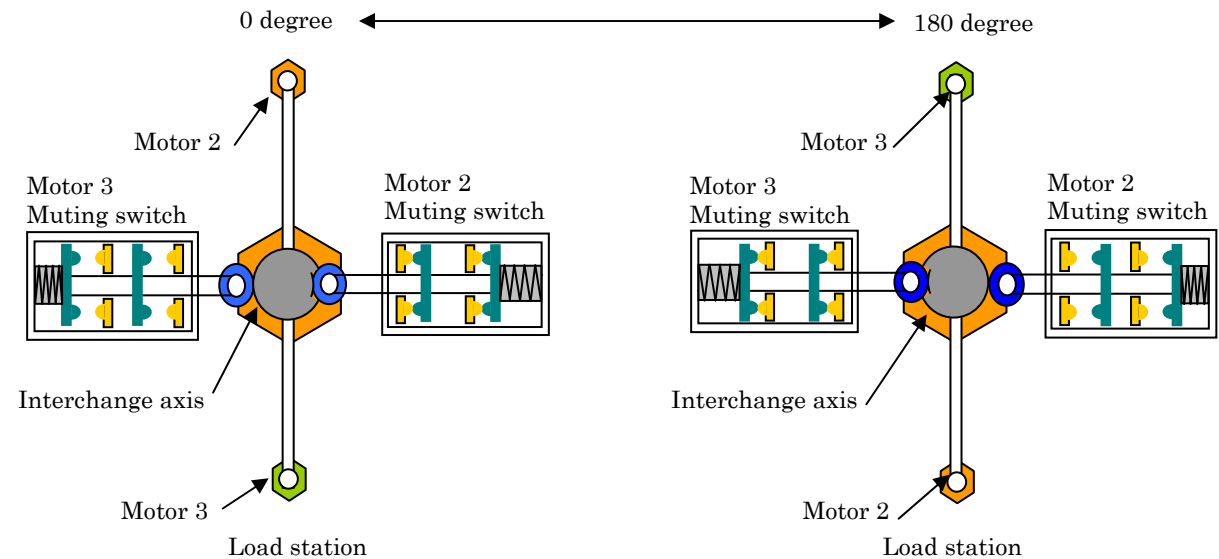
System configuraion with Ferris Wheel positioner

5.2 Muting switches

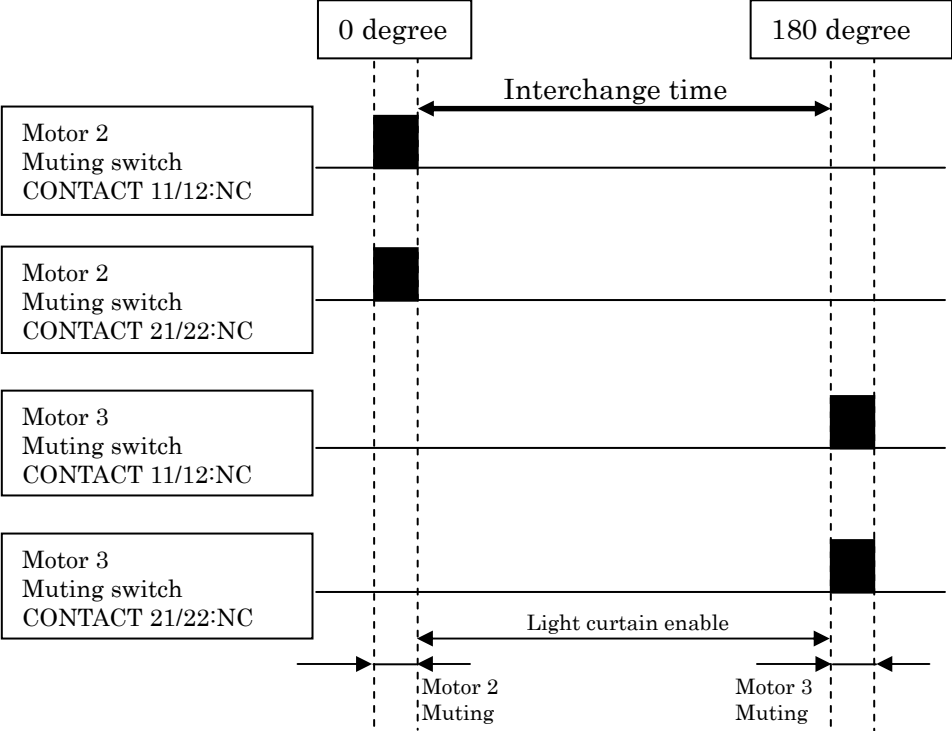
In the local stop system with a Ferris wheel, the local stop axis is changed alternately by the position of the interchange axis when the work piece is replaced at load station.

Therefore, it is necessary to installing the muting switch in the interchange axis corresponding to the position of the interchange axis.

In the following figure, the Muting switch for motor 2 mutes the local stop function of the Motor 2 when the position of interchange axis is 0 degree. On the other hand, the Muting switch for motor 3 mutes the local stop function of the Motor 3 when the position of interchange axis is 180 degree.



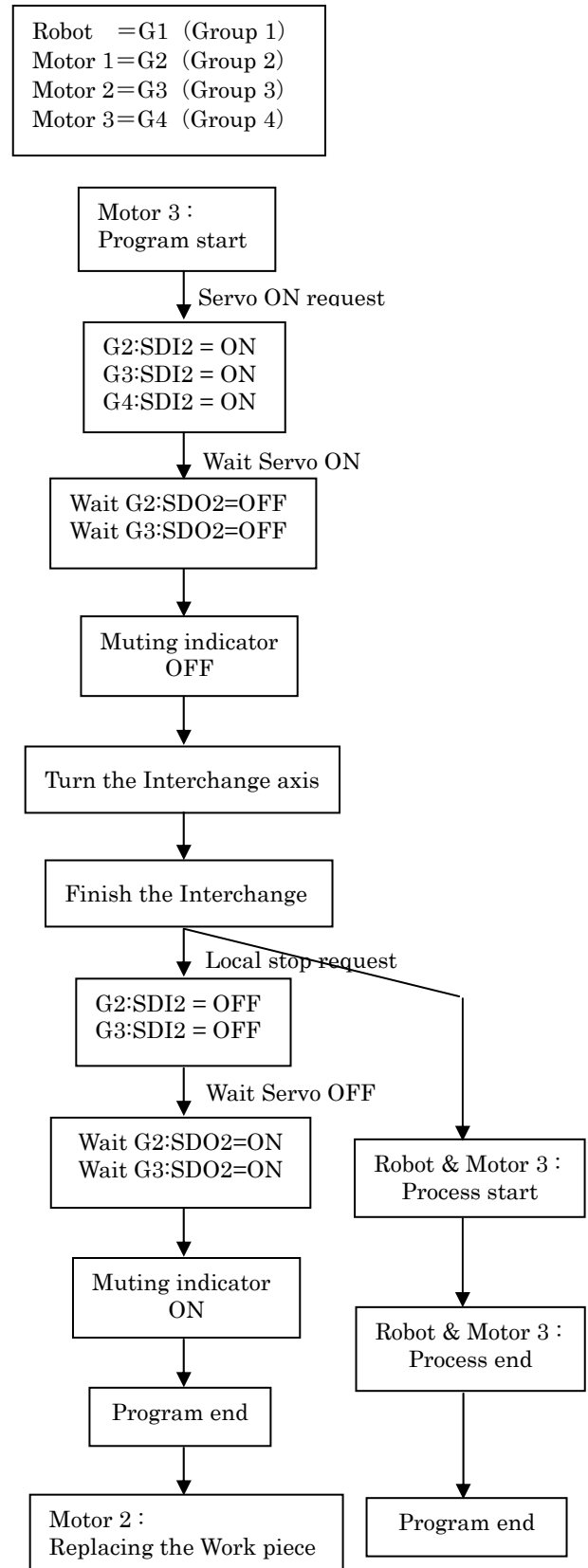
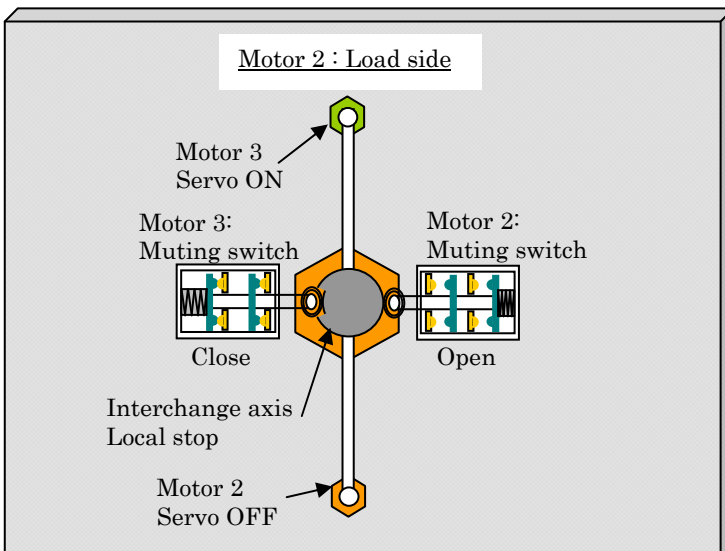
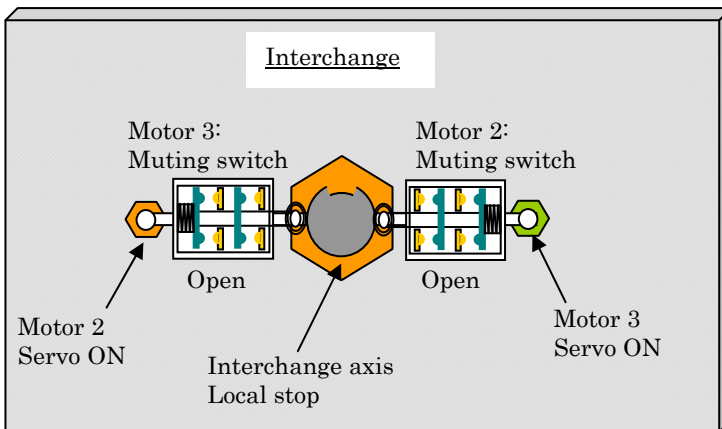
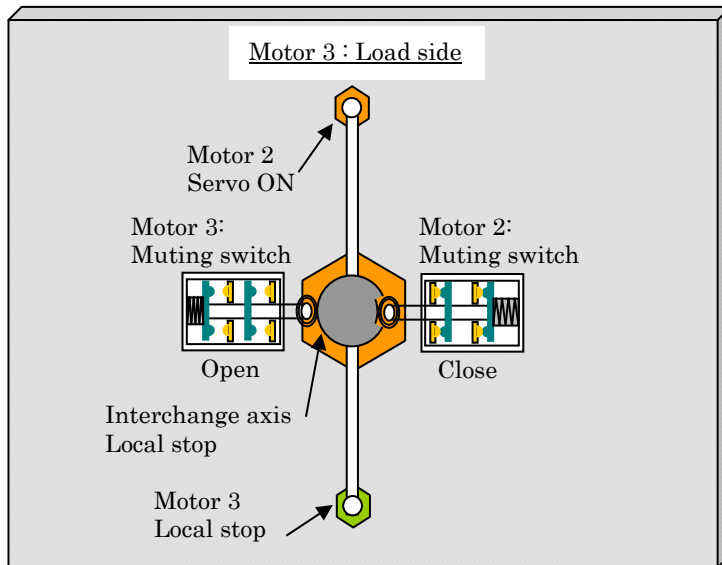
The sequence diagram of muting switches is shown as follows. The Muting switch of motor 2 is closed when the position the interchange axis is 0 degree, and the Muting switch of motor 3 is closed when the position the interchange axis is 180 degree. Direct opening action switch should be used as Muting switch for safety requirements.



The sequence of Muting switches

5.3 Flowchart of Ferris Wheel system

Flowchart of Ferris Wheel system is shown as follows. After replacing the work piece of load station, program(Motor 3) is started. After the interchange axis is turned, the processing of work piece at welding station and replacing work piece at load station is executed simultaneously. After the program is finished, program(Motor 2) is started.

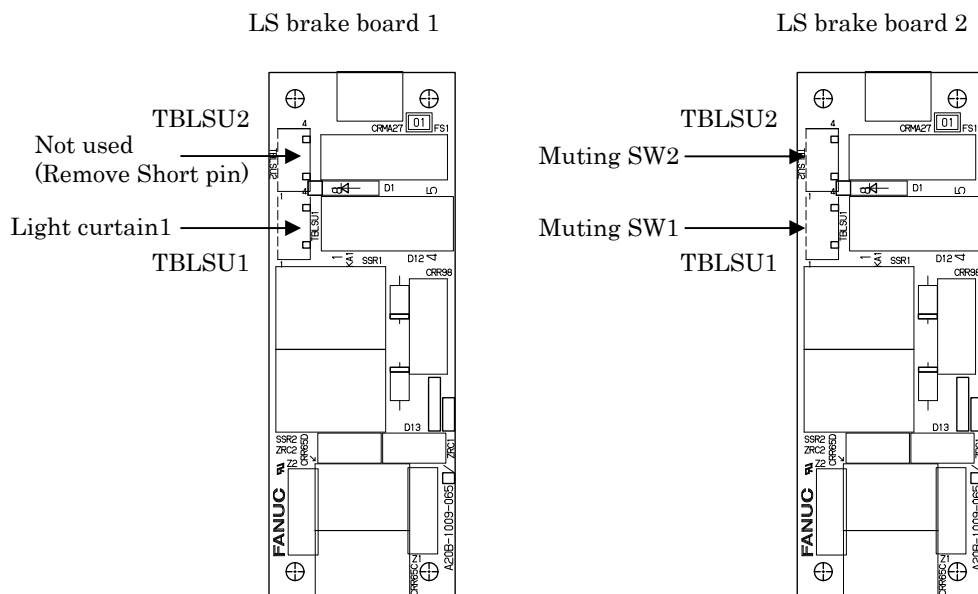


5.4 Connection of local stop function with Ferris Wheel

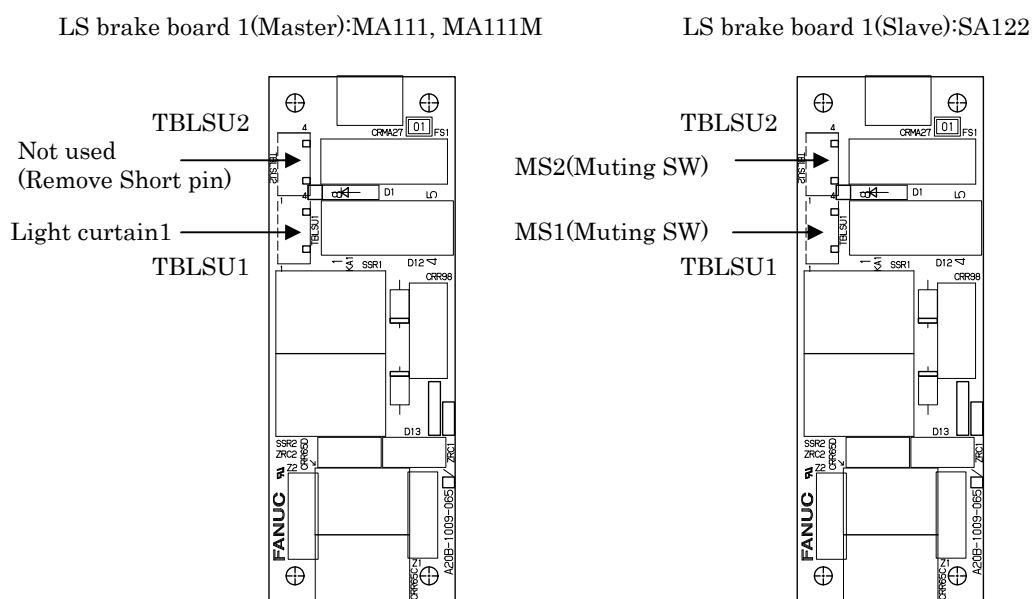
The Ferris Wheel system is used (2) LS brake board. The connection of light curtain and muting switch has to be connected to terminal(TBLSU1,TBLSU2) of LS brake board as follows.

In case of MA131, MA131M, MA133, MA133M

MB131, MB131M, MB133, MB133M, MB231, MB231M, MB233



In case of Dual arm with Ferris wheel(DA131, DA131M)



5.5 Setting of Safety I/O for local stop function with Ferris Wheel

To use the Ferris Wheel system, DCS Safe I/O connect option (A05B-2600-J568) is needed. The setting in "4.4 Setting of Safety I/O" is the initial setting. Please setup as follows when the actual muting switches are connected. The necessary change is the underlined items in FENCE column. Please note that the Safe I/O type is changed from SPI to SIR.

TYPE	No.	STO-OUT	STO-FB	FENCE	\$LS_IOPORT[n].\$SDO1_P_NUM
MA131	1	SPO[9]	SPI[13]-1	SPI[9]	\$LS_IOPORT[1].\$SDO1_P_NUM = 9
	2	SPO[11]	SPI[14]-1	<u>SIR[1]</u>	\$LS_IOPORT[2].\$SDO1_P_NUM = 11
	3	SPO[12]	SPI[14]-2	<u>SIR[2]</u>	\$LS_IOPORT[3].\$SDO1_P_NUM = 12
MA131M	1	SPO[9]	SPI[13]-2	SPI[9]	\$LS_IOPORT[1].\$SDO1_P_NUM = 9
	2	SPO[11]	SPI[14]-1	<u>SIR[1]</u>	\$LS_IOPORT[2].\$SDO1_P_NUM = 11
	3	SPO[12]	SPI[14]-2	<u>SIR[2]</u>	\$LS_IOPORT[3].\$SDO1_P_NUM = 12
DA131	1	SPO[1]	SPI[2]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[49]	SPI[58]-1	<u>SIR[1]</u>	\$LS_IOPORT[1].\$SDO1_P_NUM = 49
	3	SPO[50]	SPI[58]-2	<u>SIR[2]</u>	\$LS_IOPORT[2].\$SDO1_P_NUM = 50
DA131M	1	SPO[1]	SPI[2]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[49]	SPI[58]-1	<u>SIR[1]</u>	\$LS_IOPORT[1].\$SDO1_P_NUM = 49
	3	SPO[50]	SPI[58]-2	<u>SIR[2]</u>	\$LS_IOPORT[2].\$SDO1_P_NUM = 50
MB131	1	SPO[1]	SPI[5]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	<u>SIR[1]</u>	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[4]	SPI[6]-2	<u>SIR[2]</u>	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
MB131M	1	SPO[1]	SPI[5]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	<u>SIR[1]</u>	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[4]	SPI[6]-2	<u>SIR[2]</u>	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
MB231	1	SPO[1]	SPI[5]-1	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	<u>SIR[1]</u>	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[3]	SPI[6]-2	<u>SIR[1]</u>	
	4	SPO[4]	SPI[7]-1	<u>SIR[2]</u>	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
	5	SPO[4]	SPI[7]-2	<u>SIR[2]</u>	
MB231M	1	SPO[1]	SPI[5]-2	SPI[1]	\$LS_IOPORT[1].\$SDO1_P_NUM = 1
	2	SPO[3]	SPI[6]-1	<u>SIR[1]</u>	\$LS_IOPORT[2].\$SDO1_P_NUM = 3
	3	SPO[3]	SPI[6]-2	<u>SIR[1]</u>	
	4	SPO[4]	SPI[7]-1	<u>SIR[2]</u>	\$LS_IOPORT[3].\$SDO1_P_NUM = 4
	5	SPO[4]	SPI[7]-2	<u>SIR[2]</u>	

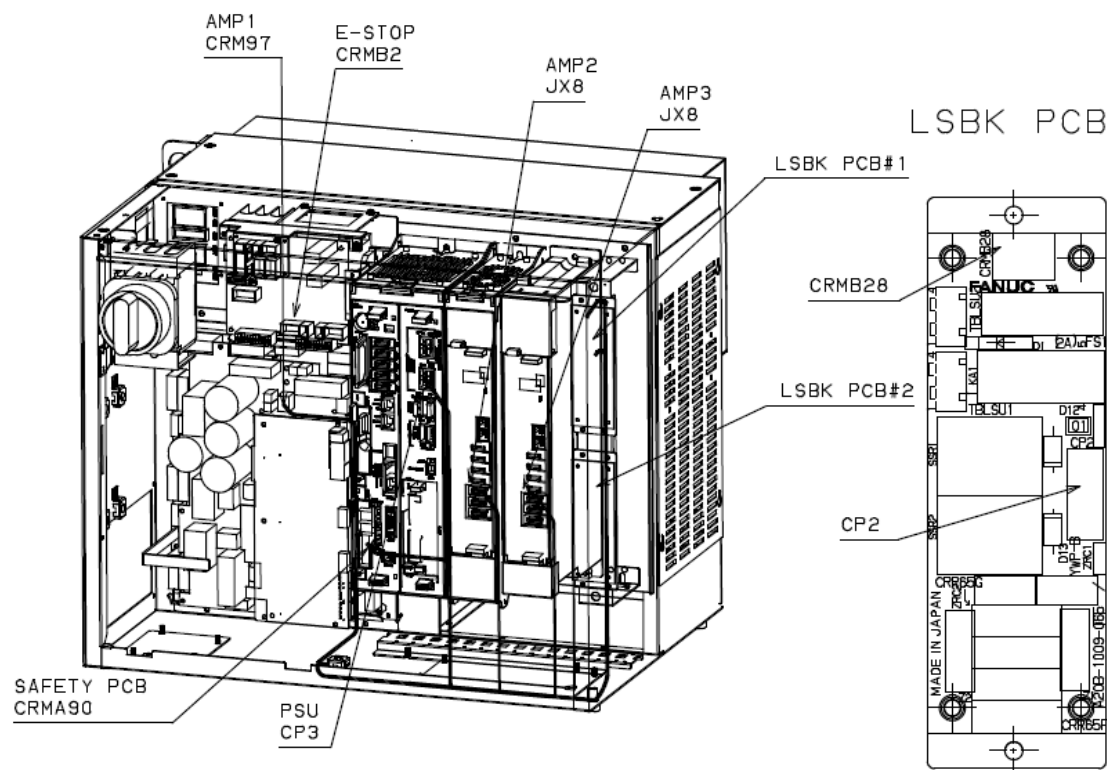
Please setup DCS Safe I/O connect as follows.

TYPE	DCS Safe I/O connect
MA131, MA131M	<u>SIR[1]</u> = SPI[9] OR SPI[11] <u>SIR[2]</u> = SPI[9] OR SPI[12]
DA131, DA131M	<u>SIR[1]</u> = SPI[1] OR SPI[49] <u>SIR[2]</u> = SPI[1] OR SPI[50]
MB131, MB131M MB231, MB231M	<u>SIR[1]</u> = SPI[1] OR SPI[3] <u>SIR[2]</u> = SPI[1] OR SPI[4]

6 Mounting and Internal Wiring

The mounting and internal wiring for local stop function with STO are shown as following.

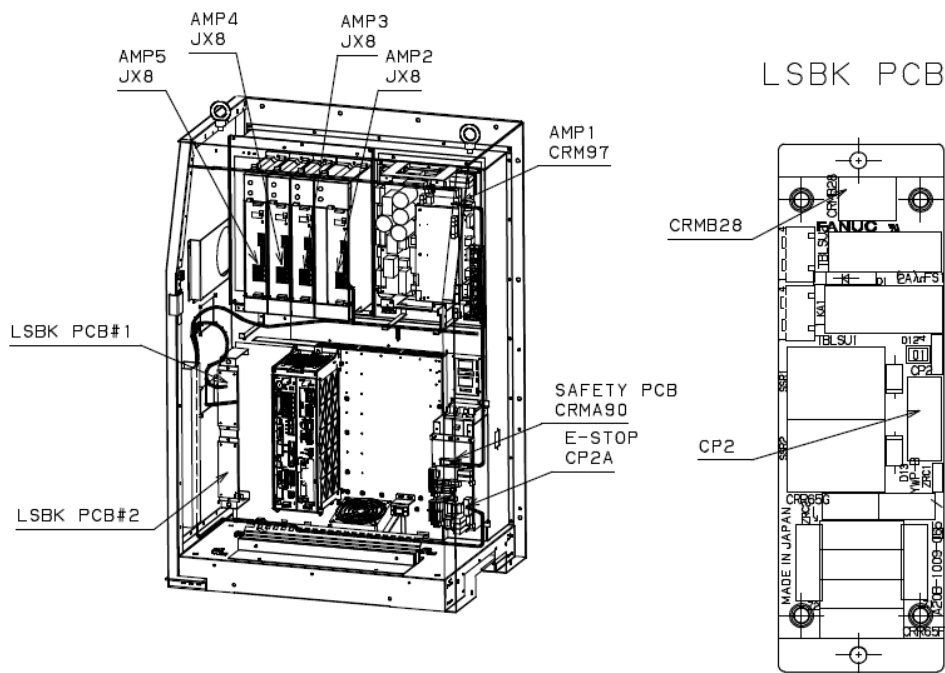
Local stop control and LS brake board(A-cabinet)



LS control	Cable specification	Connection
A05B-2606-J070	A660-4005-T354#L500R0 A660-2006-T568#L1R003F	AMP1 (CRM97) -LSBK#1 (CRMA28) -E-STOP (CRMB2) -AMPn (JX8) PSU (CP3) -LSBK#1 (CP2)
A05B-2606-J072	A660-4005-T355#L500R0 A660-2006-T568#L1R003F	AMP1 (CRM97) -LSBK#1 (CRMA28) -SAFETY PCB (CRMA90) -AMPn (JX8) PSU (CP3) -LSBK#1 (CP2)
A05B-2606-J074	A660-4005-T356#L400R0 A660-2006-T568#L1R003F	AMP1 (CRM97) -LSBK#1 (CRMA28) -SAFETY PCB (CRMA90) -AMP2, 3 (JX8) PSU (CP3) -LSBK#1 (CP2)
A05B-2606-J076	A660-4005-T361#L400R0 A660-4005-T382#L700R0A	AMP1 (CRM97) -LSBK#1, #2 (CRMA28) -SAFETY PCB (CRMA90) -AMP2, 3 (JX8) PSU (CP3) -LSBK#1, #2 (CP2)
A05B-2606-J082	A660-4005-T374#L500R0 A660-2006-T568#L1R003F	AMP1 (CRM97) -LSBK#1 (CRMA28) -E-STOP (CRMB2, TBOP13 (5, 7)) -AMPn (JX8) PSU (CP3) -LSBK#1 (CP2)

LS Brake board	Item
A05B-2605-J403	LSBK PCB (A20B-1009-0650) with (4) -M4 screws
A05B-2606-J432	Mounting Plate with (2) -M4 screws

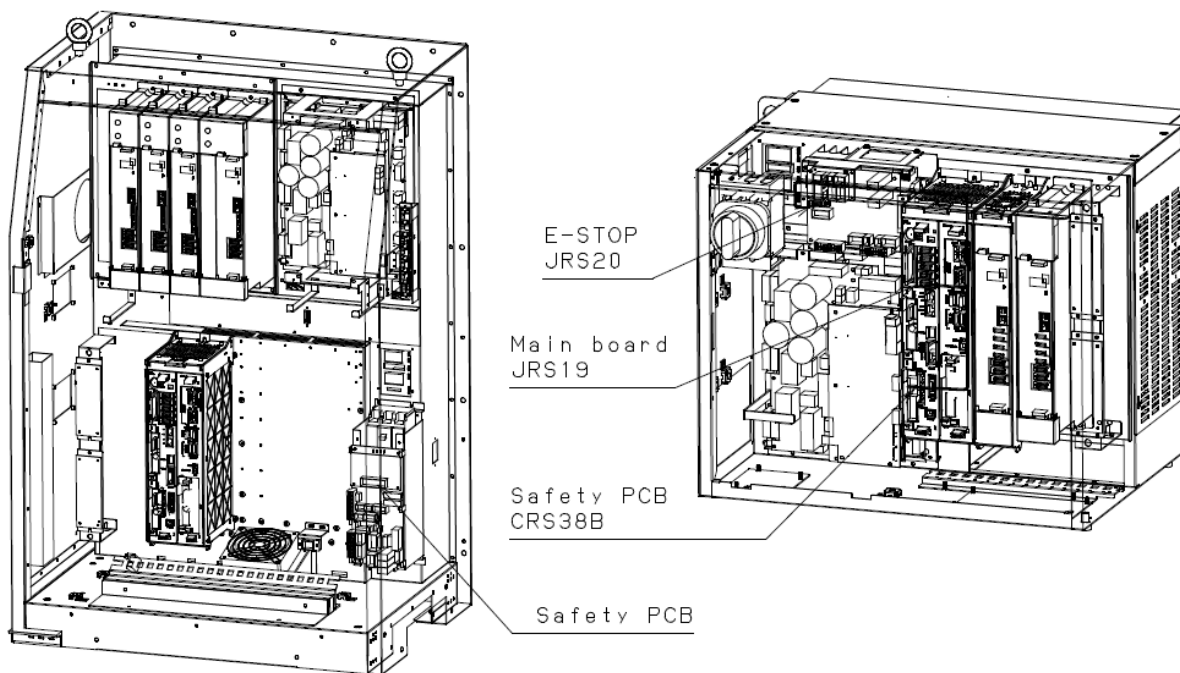
Local stop control and LS brake board(B-cabinet)



Ordering number	Cable specification	Connection
A05B-2608-J070	A660-4005-T375#L750R0 A660-2006-T568#L1R503G	AMP1 (CRM97) -LSBK#1 (CRMA28) -SAFETY PCB (CRMA90) -AMPn (JX8) E-STOP (CP2A) -LSBK#1 (CP2)
A05B-2608-J072	A660-4005-T355#L750R0B A660-2006-T568#L1R503G	AMP1 (CRM97) -LSBK#1 (CRMA28) -SAFETY PCB (CRMA90) -AMPn (JX8) E-STOP (CP2A) -LSBK#1 (CP2)
A05B-2608-J074	A660-4005-T356#L650R0B A660-2006-T568#L1R503G	AMP1 (CRM97) -LSBK#1 (CRMA28) -SAFETY PCB (CRMA90) -AMPn (JX8) PSU (CP3) -LSBK#1 (CP2)
A05B-2608-J076	A660-4005-T361#L650R0B A660-4005-T382#L1R203B	AMP1 (CRM97) -LSBK#1, #2 (CRMA28) -SAFETY PCB (CRMA90) -AMPn, n+1 (JX8) PSU (CP3) -LSBK#1, #2 (CP2)
A05B-2608-J078	A660-4005-T376#L650R0 A660-4005-T382#L1R203B	AMP1 (CRM97) -LSBK#1, #2 (CRMA28) -SAFETY PCB (CRMA90) -AMPn, n+1, n+2 (JX8) PSU (CP3) -LSBK#1, #2 (CP2)

LS Brake board	Item
A05B-2605-J403	LSBK PCB (A208-1009-0650) with (4) -M4 screws
A05B-2606-J432	Mounting Plate with (2) -M4 screws

Safety I/O board



Ordering number	specification	Connection
A05B-2600-J130 (B-cabinet)	A20B-3300-0690	Safety PCB and (2) M3 screws
A05B-2600-J131 (A-cabinet)	A20B-8201-0110	Safety PCB
A05B-2600-J200 (A-cabinet)	A660-4005-T351#L800R0	E-STOP (JRS20) -Main board (JRS19) -Safety PCB (CRS38B)

7 Circuit diagram

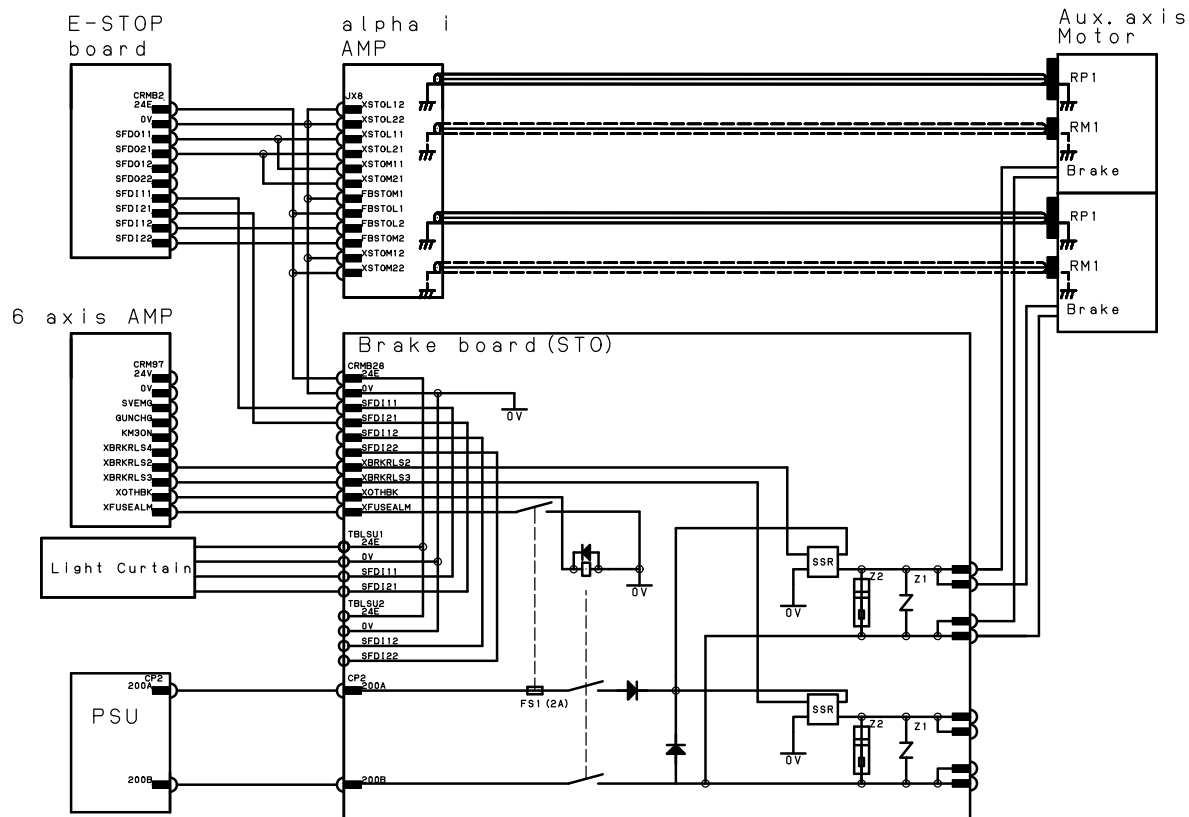
Circuit diagram of MA111 / MA111M / MA211

MA111/MA211

1 axis 1 group total 1 axis

2 axis 1 group total 2 axis

In case of MA211

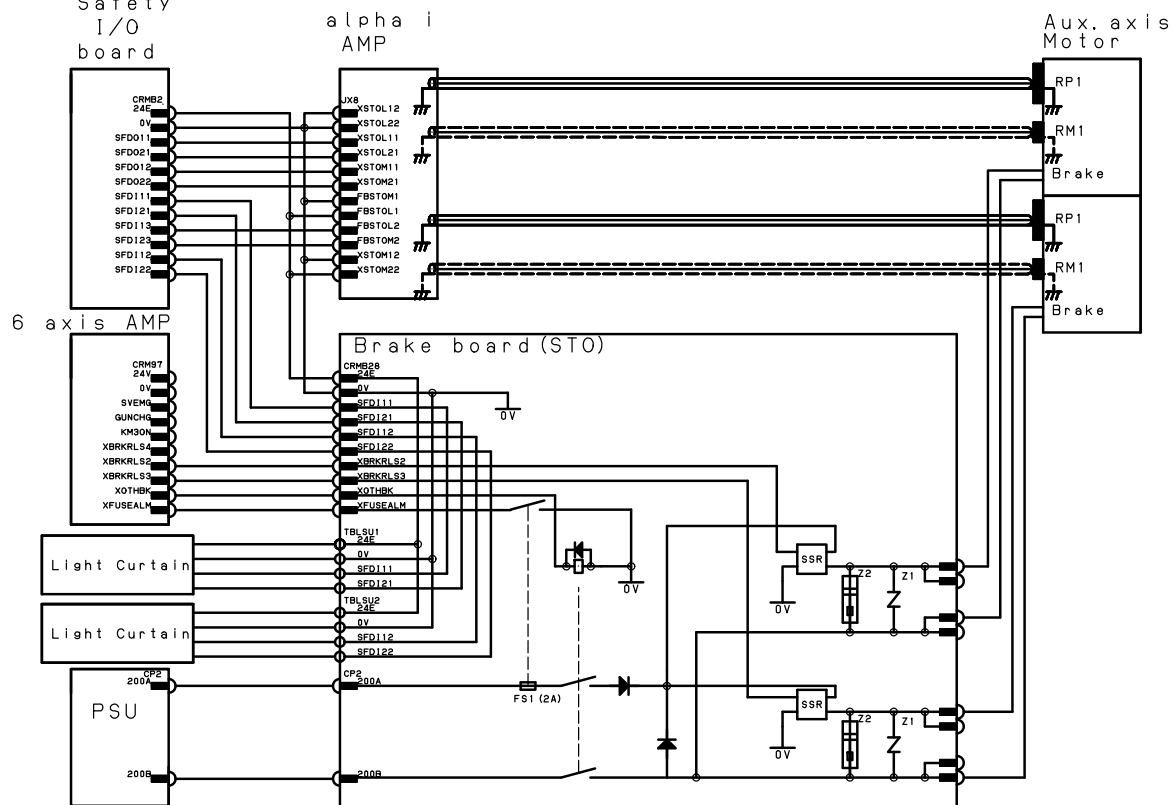


Circuit diagram of MA122 / MB122

MA122

1 axis 2 group total 2 axis

Safety
I/O
board



Circuit diagram of MA222 / MB222

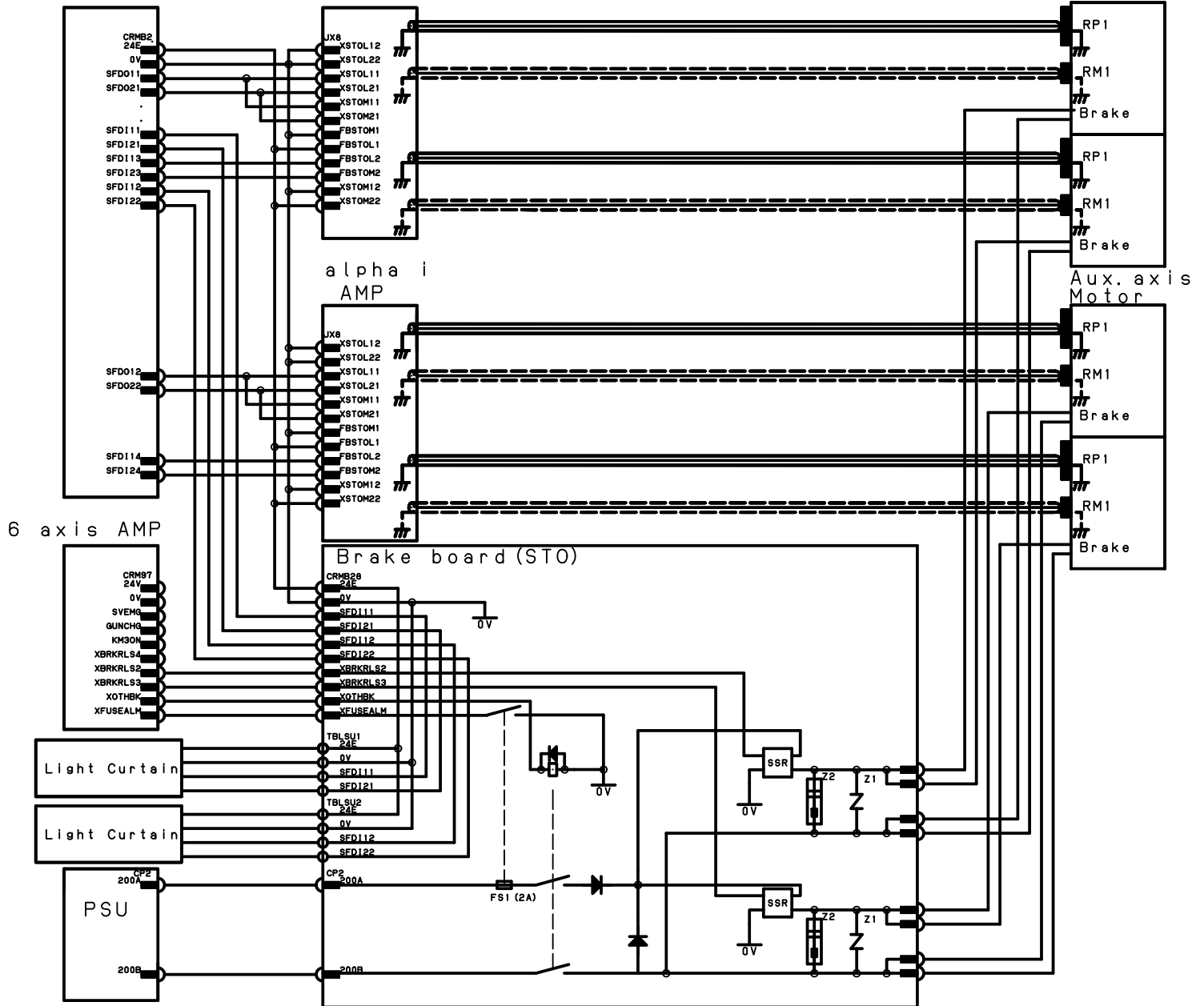
MA222

2 axis 2 group total 4 axis

Safety
I/O
board

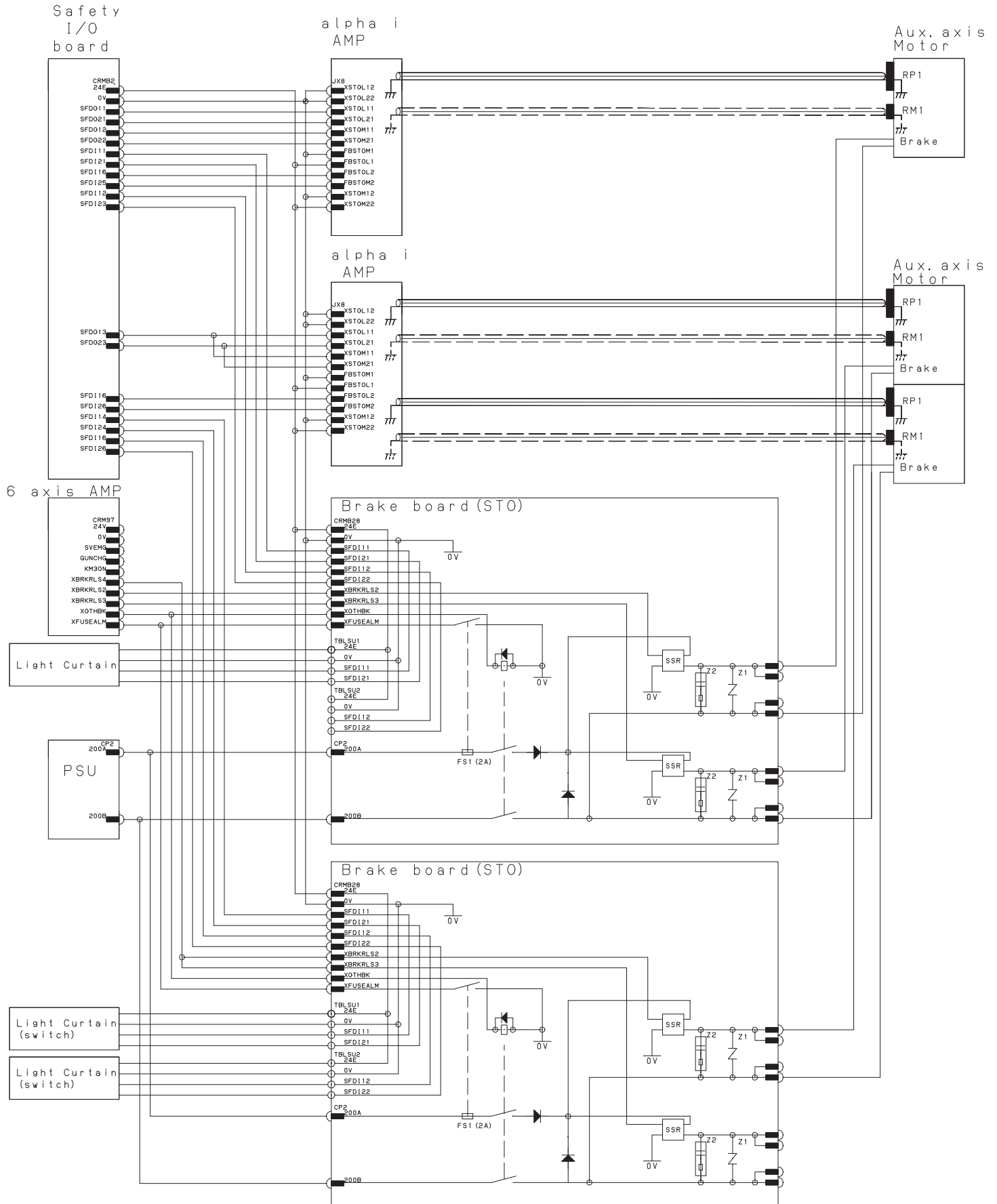
alpha i
AMP

Aux. axis
Motor



Circuit diagram of MA131 / MA131M / MA133 / MA133M /
MB131 / MB131M / MB133 / MB133M

MA131/MA133
 1 axis 3 group total 3 axis

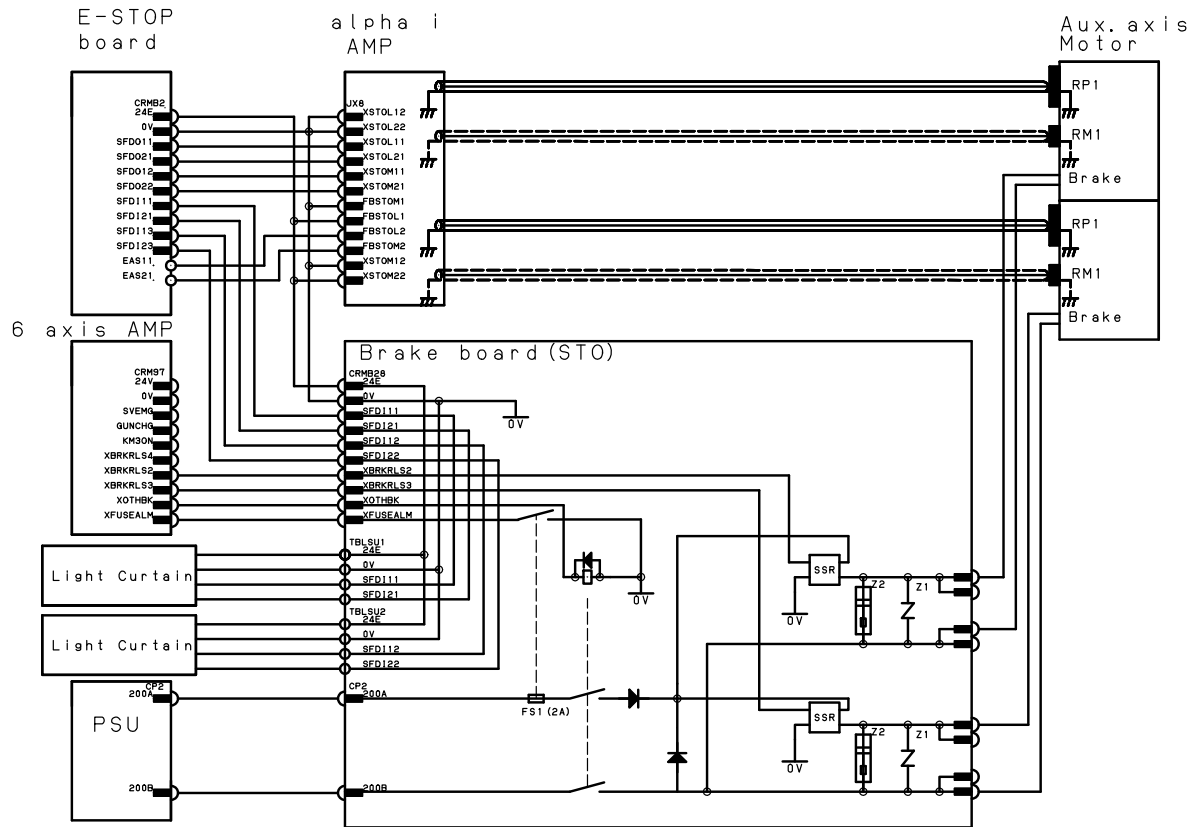


Circuit diagram of SA122

SA122

(Slave controller)

1 axis 2 group total 2 axis



Circuit diagram of MB111 / MB111M / MB211

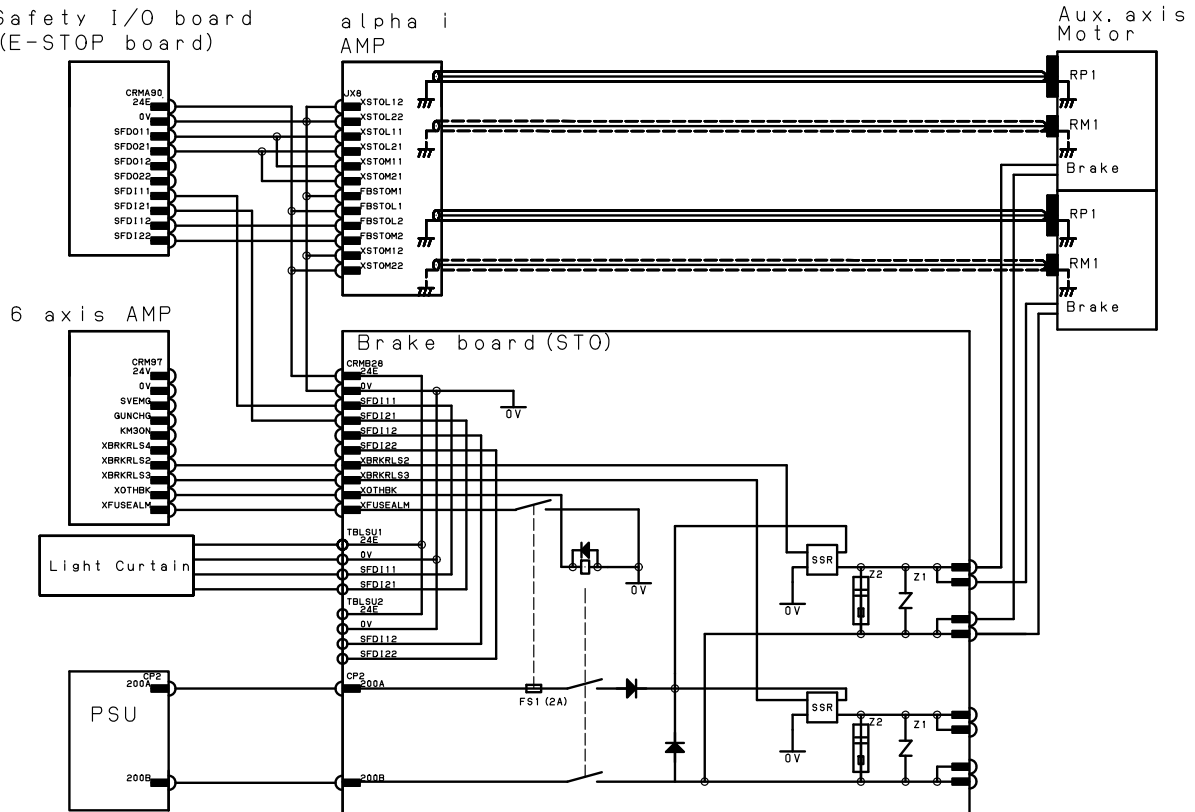
MB111/MB211

1 axis 1 group total 1 axis

2 axis 1 group total 2 axis

In case of MB211

Safety I/O board
(E-STOP board)



Circuit diagram of MB233 / MB231 / MB231M

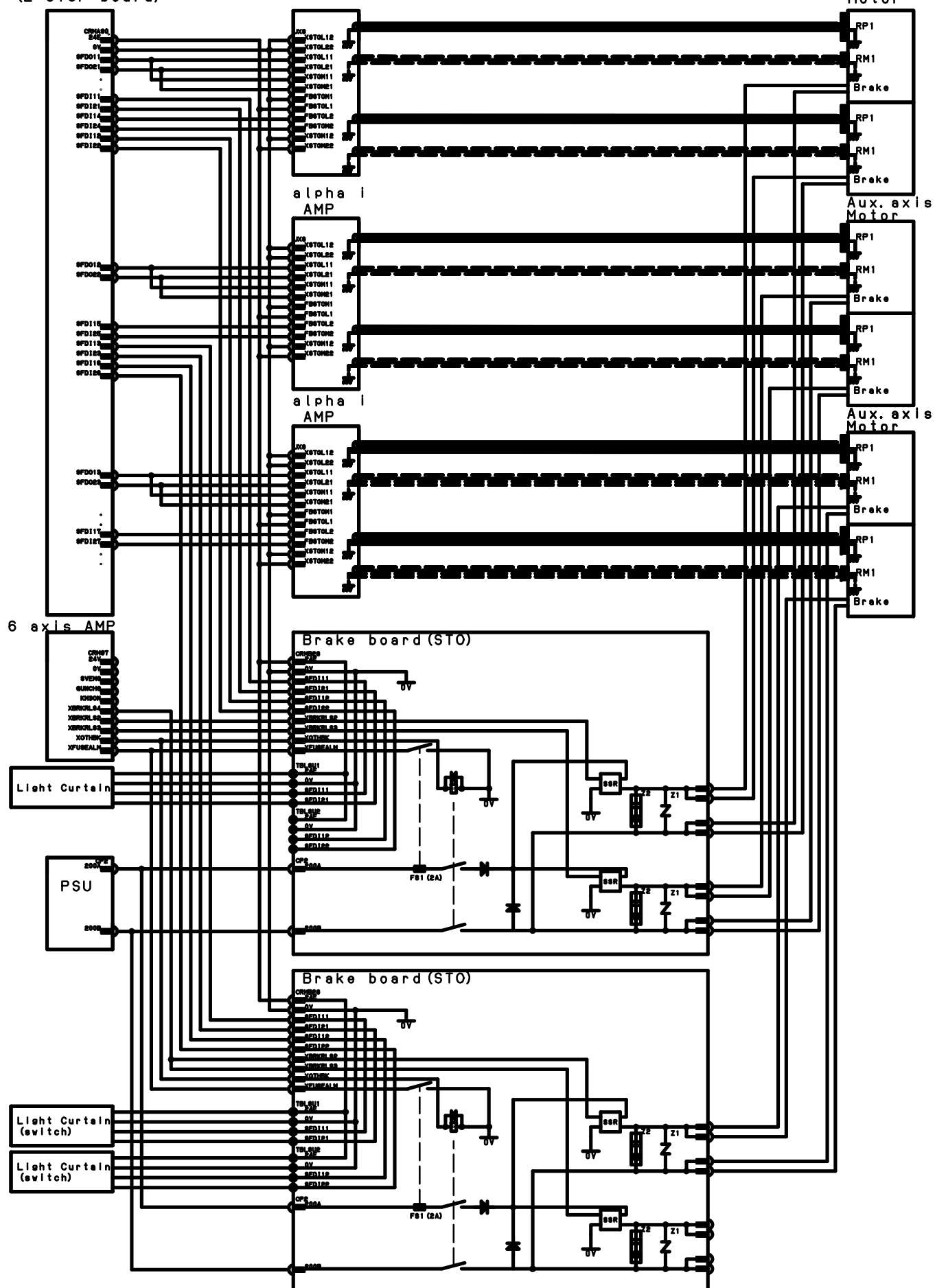
MB231/MB233

2 axis 3 group total 5 axis

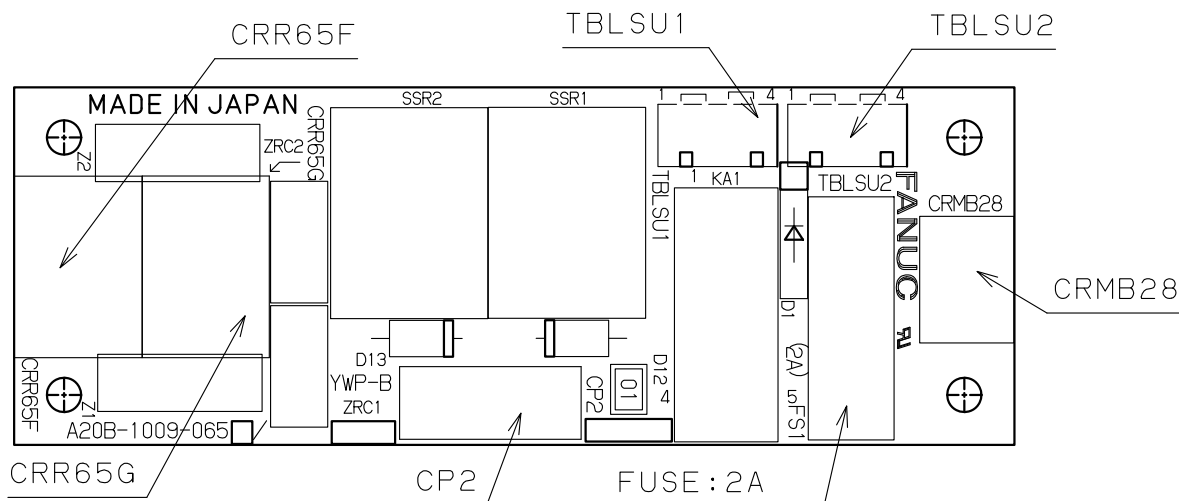
Safety I/O board
(E-STOP board)

alpha i
AMP

Aux. axis
Motor



Connector table



JX8 (AMPx)
PCR20

11	XSTOM12	01	XSTOL12
12		02	XSTOL22
13	XSTOM11	03	XSTOL11
14		04	XSTOL21
15		05	FBSTOM2
16	XSTOM22	06	
17	XSTOM21	07	FBSTOL2
18		08	FBSTOM1
19		09	
20		10	FBSTOL1

CRMA90 (SAFETY IO)
D1100 (X)

	A	B
01	24E	0V
02	SFD011	SFD021
03	SFD012	SFD022
04	SFD013	SFD023
05	SFD014	SFD024
06	SFD015	SFD025
07	SFD016	SFD026
08	SFD017	SFD027
09	SFD018	SFD028
10		
11	SFD0011	SFD0021
12	SFD0012	SFD0022
13	SFD0013	SFD0023
14	SFD0014	SFD0024
15	SFD0015	SFD0025
16	SFD0016	SFD0026
17	SFD0017	SFD0027
18	SFD0018	SFD0028
19	0V	0V
20	0V	0V

CRMB2 (E-STOP)
D1200 (Y)

A1	24E	B1	0V
A2	24E	B2	0V
A3	SFDI11	B3	SFDI21
A4	SFDI12	B4	SFDI22
A5	SFDO11	B5	SFDO21
A6	SFDO12	B6	SFDO22

CRM97 (AMP1)
D2100 (Y)

A1	BKRLS_A	B1	24V
A2	BKRLS_B	B2	0V
A3	BKRLS_C	B3	XFUSEALM
A4	GUNCHG	B4	XSVEMG
A5	KM30N	B5	OTHBK
A6		B6	

CRMB28
D2100 (X)

A1	24E	B1	0V
A2	SFDI11	B2	SFDI21
A3	SFDI12	B3	SFDI22
A4	BKRLS_A	B4	BKRLS_B
A5	AUXEMG	B5	FUSEALM
A6		B6	

CP2
(D-3200, 3p1n, X)

1	200A
2	200B
3	

TBLSU1
TERMINAL

1	24E
2	SFDI11
3	0V
4	SFDI21

TBLSU2
TERMINAL

1	24E
2	SFDI12
3	0V
4	SFDI22

CRR65F
(D-3200, 6pin, XX)

A1	BKA2	B1	BKA2
A2		B2	
A3	BKCOM	B3	BKCOM

CRR65F
(D-3200, 6pin, XX)

A1	BKA1	B1	BKA1
A2		B2	
A3	BKCOM	B3	BKCOM

8 Trouble shooting and Maintenance

8.1 Trouble shooting

(1) SRVO-215 Brake Unit fuse blown(R:%d)

SRVO-215 Fuse blown (Aux. Axis)	
[Explanation]	The fuse FUSE1 on the Brake board is blown.
[Action]	Replace this FS1(fuse) and SSR after replacing the brake of brake cable. SSR should be replaced simultaneously for the fuse because SSR may be damaged by short circuit current.

Item	Specification
Fuse and SSR kit	A05B-2606-K002

(2) SRVO-450 Drvoff circuit fail(G:%d A:%d)

SRVO-450 Drvoff circuit fail(G:%d A:%d)	
[Explanation]	The system variable (\$LS_IOPORT[x].\$MO_GRP_NUM) is not matched switch setting(SW1) of servo amplifier. The STO-OUT(STO-A, STO-B) signals are not in the same status.
[Action 1]	Check the software setting and switch setting.
[Action 2]	Check the wiring of STO signals(JX8).
[Action 3]	Replace the servo amplifier(α SV).
[Action 4]	Replace the Safety I/O board(or E-stop board).

(3) SRVO-446 DCS LS Fence open %x,%x

SRVO-446 DCS LS Fence open %x,%x	
[Explanation]	In Auxiliary axis servo off (Local stop) function, the safe I/O assigned to the fence input is OFF when the auxiliary axis is not servo off condition. The value displayed in the message shows the No. of local stop that the alarm occurs. The value is hexadecimal, and each bit is corresponded to each item in DCS local stop menu (1:No.1, 2:No.2, 4:No.3, 8:No.4, 10:No.5, 20:No.6, 40:No.7, 80:No.8).
[Action]	Set the safe I/O assigned to the fence input to ON, or set the auxiliary axis servo to off.

(4) SRVO-447 DCS LS STO-FB alarm %x,%x

SRVO-447 DCS LS STO-FB alarm %x,%x
[Explanation] In Auxiliary axis servo off (Local stop) function, mismatch of STO-OUT and STO-FB is detected. The value displayed in the message shows the No. of local stop that the alarm occurs. The value is hexadecimal, and each bit is corresponded to each item in DCS local stop menu (1:No.1, 2:No.2, 4:No.3, 8:No.4, 10:No.5, 20:No.6, 40:No.7, 80:No.8). [Action 1] Check the setting of DCS local stop menu. [Action 2] Check the connection between emergency stop board and auxiliary axis servo amplifier. [Action 3] Replace the cable between emergency stop board and auxiliary axis servo amplifier. [Action 4] Replace the emergency stop board. [Action 5] Replace the auxiliary axis servo amplifier.

(5) LSTP-015 Brake setting error %d (G:%d)

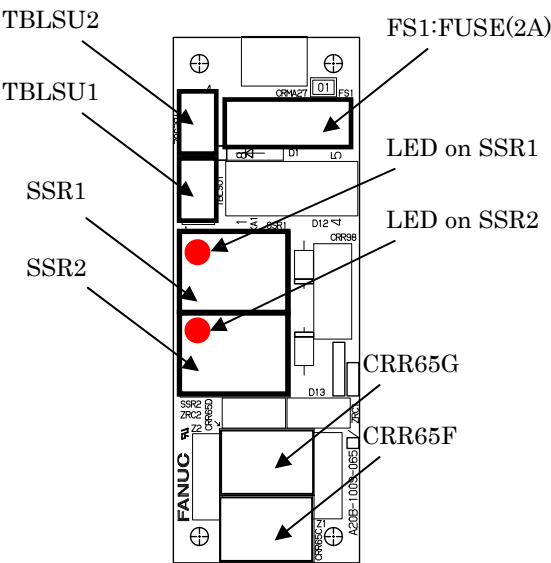
LSTP-015 Brake setting error %d (G:%d)
[Explanation] The brake number of the local stop group is zero or already used in other group. [Action] Set independent brake number for the local stop group in Auxiliary axis setup at controlled start. For correct brake number, please see “4.3 I/O Setting of each local stop control type”. It needs to cycle power to reset this alarm.

8.2 LED on LS Brake board

If LED on SSR1, SSR2 is ON, brake of motor is released.

SSR1 : CRR65F : Brake number 2(In case of MA111)

SSR2 : CRR65G : Brake number 3(In case of MA122)



9 Procedure of ordering

9.1 The ordering spec. specified by AUXILIARY AXIS ORDER MANUAL

Specify the ordering specifications according to auxiliary axis order manual.

Item	
Specify the motor specification	Select the motor for local stop axis according to this manual " <u>Application motors</u> ".
Servo card	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Servo amplifier	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Aux. Axis control	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Adapter plate of servo amplifier	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Adapter plate without fin	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
AMP Tag	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Brake cable Tag	Brake cable tag is specified according to this manual.
Gun change Tag	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Motor cable	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Gun change unit	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Battery unit	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Transformer capacity Upgrade	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Assembly instructions	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".
Spare fuse	According to the " <u>AUXILIARY AXIS ORDER MANUAL</u> ".

Afterwards, add the ordering specifications for local stop system according to the following sections.

The order for multi arm system such as DA131, DA131M, DA133 and DA133M is composed by the combination of master order and slave order as followings.

DA131(DA133) = MA111(Master) + SA122(Slave)

DA131M(DA133M) = MA111M(Master) + SA122(Slave)

9.2 Ordering specification for local stop control

Configuration No.	LS control type	Specification	Master/Slave
(1)	MA111, MA111M	A05B-2606-J070	Add to Master order
(2)	MA211	A05B-2606-J070	Add to Master order
(3)	MA122	A05B-2606-J072	Add to Master order
(4)	MA222	A05B-2606-J074	Add to Master order
(5)	MA131, MA131M	A05B-2606-J076	Add to Master order
(6)	MA133, MA133M	A05B-2606-J076	Add to Master order
(7)	SA122	A05B-2606-J082	Add to Slave order
(10)	MB111, MB111M	A05B-2608-J070	Add to Master order
(11)	MB211	A05B-2608-J070	Add to Master order
(12)	MB122	A05B-2608-J072	Add to Master order
(13)	MB222	A05B-2608-J074	Add to Master order
(14)	MB131, MB131M	A05B-2608-J076	Add to Master order
(15)	MB133, MB133M	A05B-2608-J076	Add to Master order
(16)	MB233	A05B-2608-J078	Add to Master order
(17)	MB231, MB231M	A05B-2608-J078	Add to Master order

9.3 LS Brake board

In case of A-cabinet

Item	Specification	Configurations No.									
		(1)	(2)	(3)	(4)	(5)	(6)	(7)			
LS Brake board	A05B-2605-J403	1	1	1	1	2	2	1			
LS Brake board mounting plate	A05B-2606-J432	1	1	1	1	1	1	1			

In case of B-cabinet

Item	Specification	Configurations No.									
		(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)		
LS Brake board	A05B-2605-J403	1	1	1	1	2	2	2	2		
LS Brake board mounting plate	A05B-2606-J432	1	1	1	1	1	1	1	1		

9.4 Safety I/O board

In case of A-cabinet

Item	Specification	Configurations No.									
		(1)	(2)	(3)	(4)	(5)	(6)	(7)			
Safety I/O board	A05B-2600-J131			1	1	1	1				
Cable for safety I/O board	A05B-2602-J200			1	1	1	1				

In case of B-cabinet

Item	Specification	Configurations No.									
		(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)		
Safety I/O board	A05B-2600-J130	1	1	1	1	1	1	1	1		

9.5 Brake tag for local stop

In case of A-cabinet

Item	Specification	Configurations No.									
		(1)	(2)	(3)	(4)	(5)	(6)	(7)			
BRAKE TAG(CRR65F#1)	A05B-2606-J065	1	2	1	2	1	1	1			
BRAKE TAG(CRR65G#1)	A05B-2606-J066			1	2	1	1	1			
BRAKE TAG(CRR65F#2)	A05B-2606-J067					1	1				
Cable Assembly CRR65F1	SM-C/05J***A**/65F1	1	1	1	1	1	1	1			
Cable Assembly CRR65F1	SM-C/05J***A**/65F1		1		1						
Cable Assembly CRR65G1	SM-C/05J***A**/65G1			1	1	1	1	1			
Cable Assembly CRR65G1	SM-C/05J***A**/65G1				1						
Cable Assembly CRR65F2	SM-C/05J***A**/65F2					1	1				

05J*** is specified by motor cable.

In case of B-cabinet

Item	Specification	Configurations No.									
		(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)		
BRAKE TAG(CRR65F#1)	A05B-2606-J065	1	2	1	2	1	1	2	2		
BRAKE TAG(CRR65G#1)	A05B-2606-J066			1	2	1	1	2	2		
BRAKE TAG(CRR65F#2)	A05B-2606-J066				2	1	1	2	1		
Cable Assembly CRR65F1	SM-C/07J***A**/65F1	1	1	1	1	1	1	1	1		
Cable Assembly CRR65F1	SM-C/07J***A**/65F1		1		1			1	1		
Cable Assembly CRR65G1	SM-C/07J***A**/65G1			1	1	1	1	1	1		
Cable Assembly CRR65G1	SM-C/07J***A**/65G1				1			1	1		
Cable Assembly CRR65F2	SM-C/07J***A**/65F2					1	1	1	1		
Cable Assembly CRR65F2	SM-C/07J***A**/65F2					1	1	1			

07J*** is specified by motor cable.

9.6 LS AMP TAG

In case of A-cabinet

Item	Specification	Configurations No.									
		(1)	(2)	(3)	(4)	(5)	(6)	(7)			
LS AMP TAG	A05B-2502-J290#AMP2	1	1	1				1			
LS AMP TAG	A05B-2502-J290#AMP3										
LS AMP TAG Assembly	SM-L1/06J***AMP2	1	1	1				1			
LS AMP TAG Assembly	SM-L2/06J***AMP3										

06J*** is specification of Local stop control.

In case of B-cabinet

Item	Specification	Configurations No.								
		(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	
LS AMP TAG	A05B-2502-J290#AMPn	1	1	1	1	1	1	1	1	
LS AMP TAG	A05B-2502-J290#AMPn+1				1	1	1	1	1	
LS AMP TAG	A05B-2502-J290#AMPn+2							1	1	
LS AMP TAG Assembly	SM-L1/08J***/AMPn	1	1	1	1	1	1	1	1	
LS AMP TAG Assembly	SM-L2/08J***/AMPn+1				1	1	1	1	1	
LS AMP TAG Assembly	SM-L3/08J***/AMPn+2							1	1	

08J*** is specification of Local stop control.

AMPn,AMPn+1 and AMPn+2 are the number of servo amplifier(ex. AMP2, AMP3, AMP4).

9.7 Distribution cable for Gun change unit

In case of Gun change unit is used with local stop function, distribution cable for gun change unit has to specified.

In case of A-cabinet

Item	Specification	Configurations No.								
		(1)	(2)	(3)	(4)	(5)	(6)	(7)		
Distribution cable for Gun change unit	A05B-2605-J404	1	1	1	1	1	1	1		

In case of B-cabinet

Item	Specification	Configurations No.								
		(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	
Distribution cable for Gun change unit	TBD	1	1	1	1	1	1	1	1	

9.8 Spare fuse and Spare parts

In case of A-cabinet

Item	Specification	Configurations No.								
		(1)	(2)	(3)	(4)	(5)	(6)	(7)		
Spare fuse and SSR kit	A05B-2606-K002	1	1	1	1	1	1	1		

In case of B-cabinet

Item	Specification	Configurations No.								
		(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	
Spare fuse and SSR kit	A05B-2606-K002	1	1	1	1	1	1	1	1	

Spare parts

The spare parts specification is shown below. Specify the following spare parts if necessary.

Item	Specification.	Board specification
Local stop board	A05B-2605-J403	A20B-1009-0650

9.9 Software

The software specification is shown below.

For ferris wheel positioner, "DCS Safe I/O connect function" (A05B-2600-J568) is needed.

TYPE	Item	Specification.
MA111	Extended axes Servo off (local stop) MA111	A05B-2600-J806#MA111
MA111M	Extended axes Servo off (local stop) MA111M	A05B-2600-J806#MA111M
MA211	Extended axes Servo off (local stop) MA211	A05B-2600-J806#MA211
MA122	Extended axes Servo off (local stop) MA122	A05B-2600-J806#MA122
MA222	Extended axes Servo off (local stop) MA222	A05B-2600-J806#MA222
MA131	Extended axes Servo off (local stop) MA131	A05B-2600-J806#MA131
	DCS Safe I/O connect function	A05B-2600-J568
MA131M	Extended axes Servo off (local stop) MA131M	A05B-2600-J806#MA131M
	DCS Safe I/O connect function	A05B-2600-J568
MA133	Extended axes Servo off (local stop) MA133	A05B-2600-J806#MA133
MA133M	Extended axes Servo off (local stop) MA133M	A05B-2600-J806#MA133M
SA122	Extended axes Servo off (local stop) SA122	A05B-2600-J806#SA122
DA131	Extended axes Servo off (local stop) DA131	A05B-2600-J806#DA131
	DCS Safe I/O connect function	A05B-2600-J568
DA131M	Extended axes Servo off (local stop) DA131M	A05B-2600-J806#DA131M
	DCS Safe I/O connect function	A05B-2600-J568
DA133	Extended axes Servo off (local stop) DA133	A05B-2600-J806#DA133
DA133M	Extended axes Servo off (local stop) DA133M	A05B-2600-J806#DA133M
MB111	Extended axes Servo off (local stop) MB111	A05B-2600-J806#MB111
MB111M	Extended axes Servo off (local stop) MB111M	A05B-2600-J806#MB111M
MB211	Extended axes Servo off (local stop) MB211	A05B-2600-J806#MB211
MB122	Extended axes Servo off (local stop) MB122	A05B-2600-J806#MB122
MB222	Extended axes Servo off (local stop) MB222	A05B-2600-J806#MB222
MB131	Extended axes Servo off (local stop) MB131	A05B-2600-J806#MB131
	DCS Safe I/O connect function	A05B-2600-J568
MB131M	Extended axes Servo off (local stop) MB131M	A05B-2600-J806#MB131M
	DCS Safe I/O connect function	A05B-2600-J568
MB133	Extended axes Servo off (local stop) MB133	A05B-2600-J806#MB133
MB133M	Extended axes Servo off (local stop) MB133M	A05B-2600-J806#MB133M
MB233	Extended axes Servo off (local stop) MB233	A05B-2600-J806#MB233
MB231	Extended axes Servo off (local stop) MB231	A05B-2600-J806#MB231
	DCS Safe I/O connect function	A05B-2600-J568
MB231M	Extended axes Servo off (local stop) MB231M	A05B-2600-J806#MB231M
	DCS Safe I/O connect function	A05B-2600-J568

9.10 The example of ordering specification with local stop system

Example of the ordering specification for (2) axes (2) group local stop system(MA222) with two set of (2) axis positioner is as follows.

Example order for MA222

Selection item	Specification	Amount
Adapter Plate(60mm(AMP2)+60mm(AMP3))	A05B-2605-J432	1
Aux. axis control with resistor discharge	A05B-2605-J001	1
Add Aux. axis(60mm(AMP2)+60mm(AMP3))	A05B-2605-J002	1
Servo AMP(Alpha iSV 80/80)	A05B-2605-J040#H209	2
Servo AMP(Alpha iSV 80/80)mount at AMP2	SM-A/A2/H209	1
Servo AMP(Alpha iSV 80/80)mount at AMP3	SM-A/A3/H209	1
Brake Cable Tag(CRR65C)	A05B-2605-J061	No need
Brake Cable Tag(CRR65D)	A05B-2605-J062	No need
CABLE TAG(AMP2L)	A05B-2605-J070	1
CABLE TAG(AMP2M)	A05B-2605-J071	1
CABLE TAG(AMP3L)	A05B-2605-J073	1
CABLE TAG(AMP3M)	A05B-2605-J074	1
PM/PP CABLE/STD/non-Flex/7m	A05B-2605-J160	2
LS control type MA222	A05B-2606-J074	1
LS AMP Tag		No need
LS AMP Tag assembly		No need
LS Brake board	A05B-2605-J403	1
LS Brake board mounting plate	A05B-2606-J432	1
Safety I/O board	A05B-2600-J131	1
Cable for safety I/O board	A05B-2602-J200	1
CABLE TAG(CRR65F#1)	A05B-2606-J065	2
CABLE TAG(CRR65G#1)	A05B-2606-J066	2
Cable connect to AMP2L	SM-C1/05J160/A2L/65F1	1
Cable connect to AMP2M	SM-C2/05J160/A2M/65F1	1
Cable connect to AMP3L	SM-C1/05J160/A3L/65G1	1
Cable connect to AMP3M	SM-C2/05J160/A3M/65G1	1
Extended axes Servo off (local stop) MA222	A05B-2600-J806#MA222	1